



الجمعية العلمية الطلابية
كلية الهندسة التكنولوجية | جامعة البلقاء التطبيقية

2018

ممثل

قسم هندسة الطاقة الكهربائية

مصطفى القرشي

سنوات دوائر كهربائية ١

سيركت 1

امتحانات الفينل

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طلاب البور والاتصالات



spark team



spark-fet.com





Q1: Find the voltage (v) for the network shown in Fig 1 using superposition theorem. (10marks)

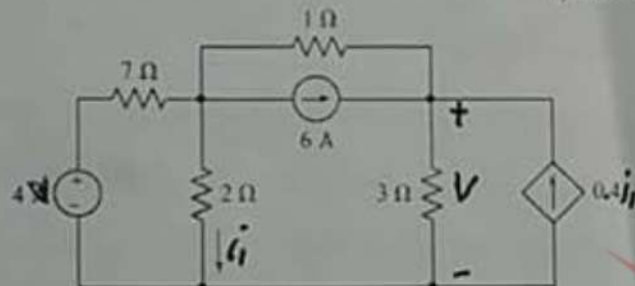


Fig 1

Q2: For the circuit shown in Fig 2:

- Determine the value of R_L to which a maximum power can be delivered.
- The maximum power delivered to R_L .

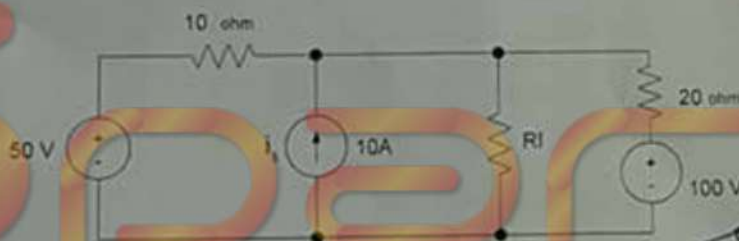


Fig 2

Q3: a) For the circuit shown in Fig 3.a find the equivalent capacitance (10 marks)

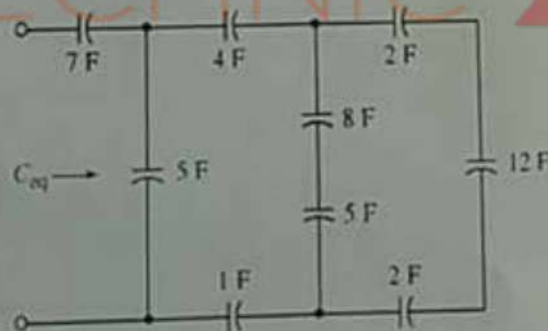


Fig 3.a

b) Calculate the power dissipated in the $40\ \Omega$ resistor and the voltage labeled V_C for the circuit given in fig 3.b

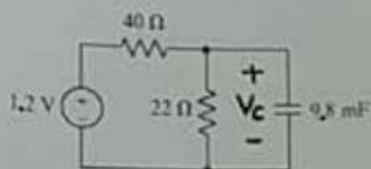


fig 3.b

Q4: The voltage across a $33\ \text{mF}$ capacitor is shown graphically in Fig 4 : (8 marks)

a) Assuming the passive sign convention, sketch the resulting current waveform across the capacitor.

b) Find the energy stored in the capacitor at $t=20\text{ms}$.

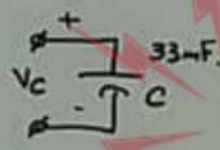
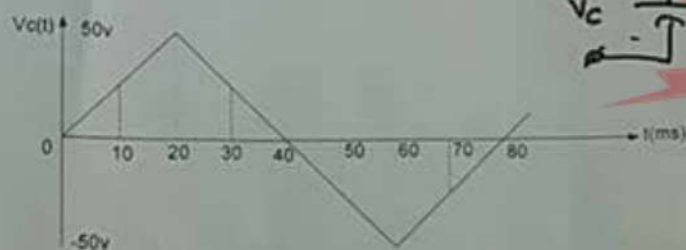


Fig 4

Q5: For the circuit shown in Fig 5 the switch opens at $t=0$ after having been closed for a long time. Find: (12 marks)

- $i_1(t)$ for $t \geq 0^-$.
- $i_L(t)$ for $t \geq 0^+$.
- $V_2(t)$ for $t \geq 0^+$.

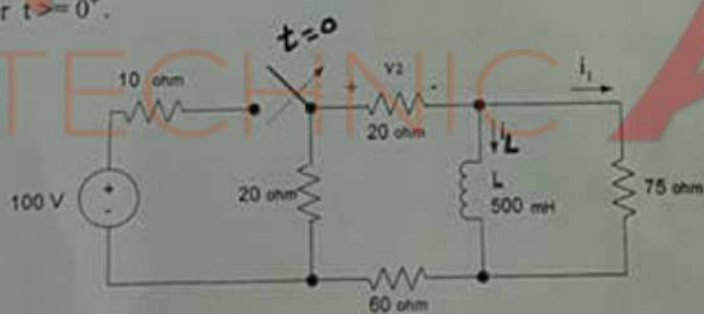


Fig 5

Good Luck

Q1

$$V = V' + V''$$

$$= 0.69 + 2.682$$

$$V = 3.372 \text{ Volts}$$

at node (W_1)

$$\frac{4 - V_1}{7} = \frac{V_1}{2} + V_1 - V_2$$

$$\rightarrow 23V_1 - 14V_2 = 8 \quad \text{--- (1)}$$

at node (W_2)

$$V_1 - V_2 + 0.4 \frac{V_1}{2} = \frac{V_2}{3} \rightarrow 7.2V_1 - 8V_2 = 0 \quad \text{--- (2)}$$

$$V_1 = 0.76 \text{ Volts}$$

at mesh (1)

$$I_1 = 0.4 \text{ A} = 0.4(I_2 - I_3)$$

at mesh (2)

$$6I_2 - 2I_3 - 3I_1 = -6$$

$$6I_2 - 2I_3 - 3(0.4(I_2 - I_3)) = -6$$

$$4.8I_2 - 0.8I_3 = -6$$

at mesh (3)

$$-2I_2 + 9I_1 = 0$$

$$I_2 = -1.298 \text{ A}$$

$$I_3 = -0.288 \text{ A}$$

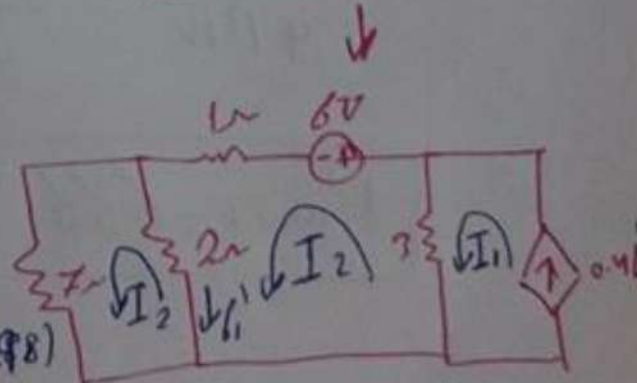
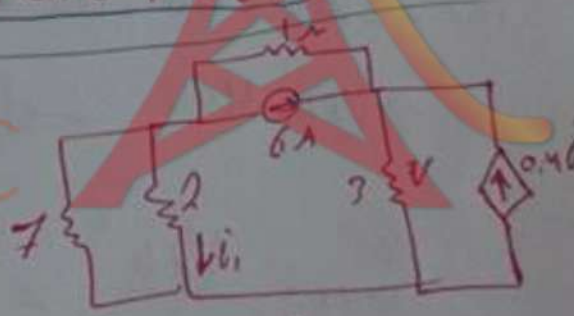
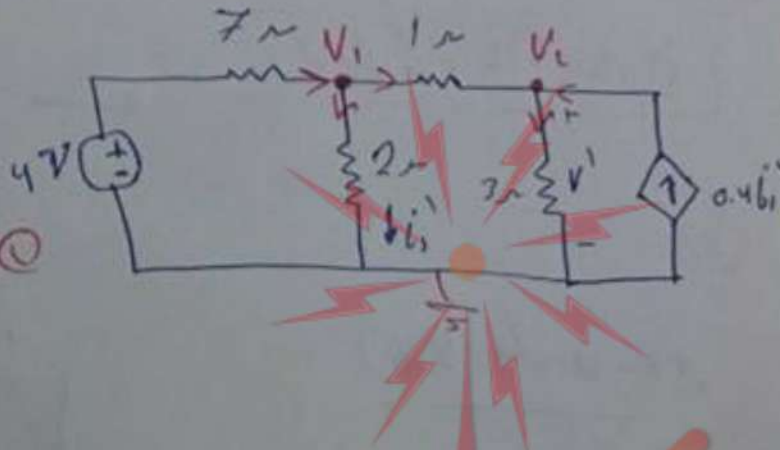
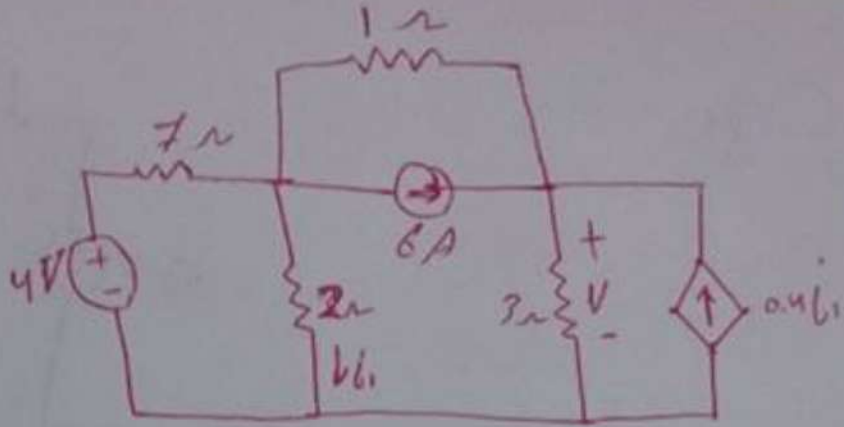
$$V'' = 7(I_1 - I_2)$$

$$V'' = 7(-0.404 + 1.298)$$

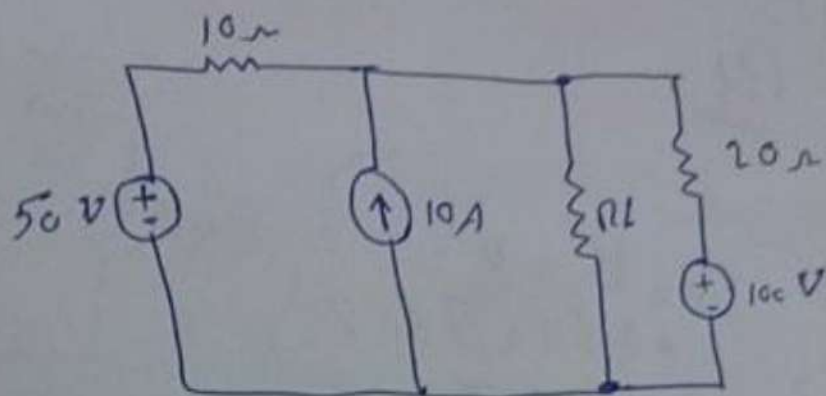
$$V'' = 2.682 \text{ Volts}$$

$$I_1 = 0.4(-1.298 + 0.288)$$

$$I_1 = -0.404$$

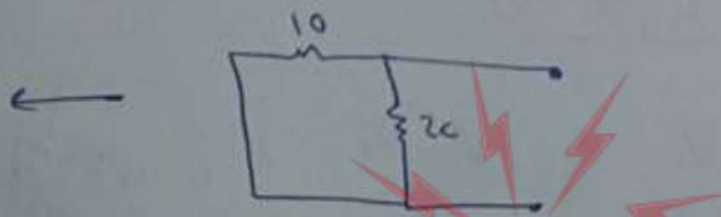


Q2



a)

$$R_{th} = R_L = 6.667 \Omega$$



b)

at node (V_{th})

$$\frac{50 - V_{th}}{10} + 10 + \frac{100 - V_{th}}{20} = 0$$

$$100 - 2V_{th} + 200 + 100 - V_{th} = 0$$

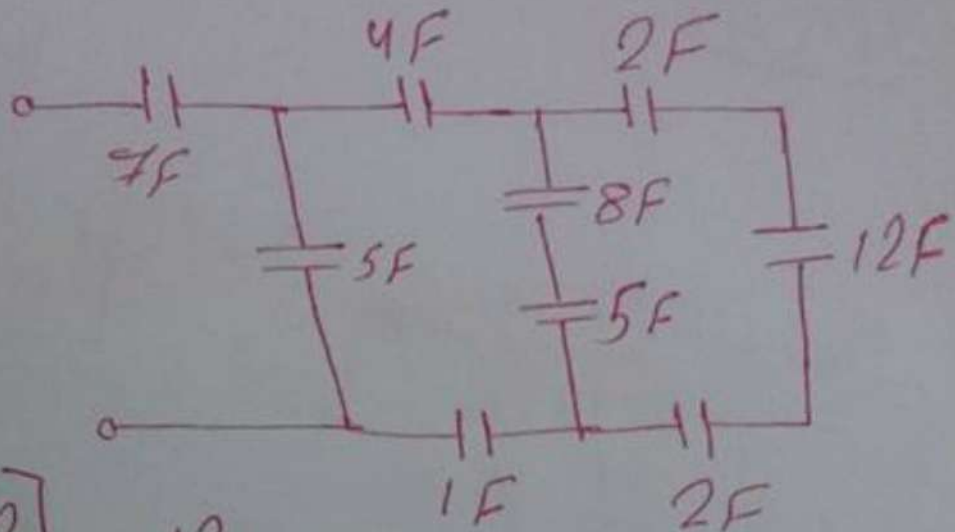
$$3V_{th} = 400$$

$$\rightarrow V_{th} = 133.34 \text{ Volts}$$

$$P_{max} = \frac{V_{th}^2}{4 R_{th}} = \frac{(133.34)^2}{4 \times 6.667} = 666.7 \text{ watt}$$

$$P_{max} = 666.7 \text{ watt}$$

Q3 (a)



$$[2 \text{ series } 12 \text{ series } 2] \rightarrow \frac{12}{13}$$

$$[8 \text{ series } 5] \rightarrow \frac{40}{13}$$

$$\left[\frac{12}{13} \parallel \frac{40}{13} \right] \rightarrow 4$$

$$[4 \text{ series } 4 \text{ series } 1] \rightarrow \frac{2}{3}$$

$$\left[\frac{2}{3} \parallel 5 \right] \rightarrow \frac{17}{3}$$

$$C_{eq} \Rightarrow \left[\frac{17}{3} \text{ series } 7 \right]$$

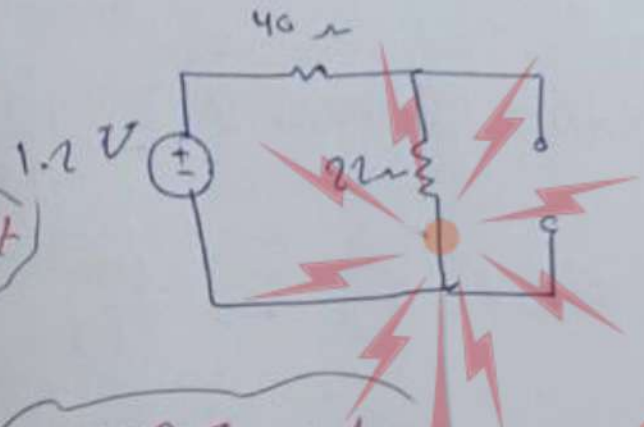
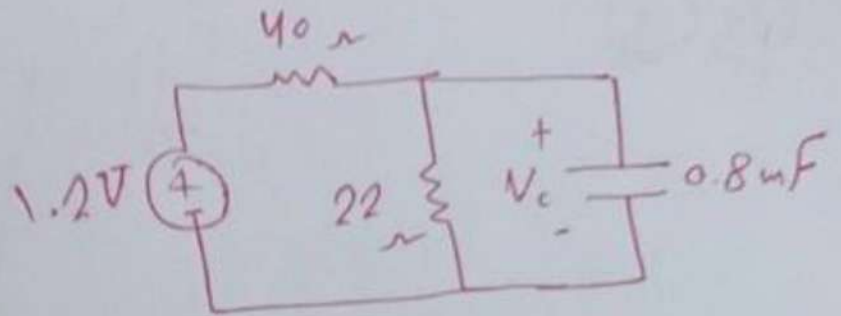
$$C_{eq} = 3.13 \text{ F}$$



Q3 (b)

$$V_{40\Omega} = \frac{1.2 \times 40}{40 + 22} = \frac{24}{31} \text{ Volt}$$

$$P_{40} = \frac{V^2}{R} = 0.0149 \text{ watt}$$



$$V_c = V_{22\Omega} = \frac{1.2 \times 22}{22 + 40} = 0.425 \text{ Volt}$$

POLYTECHNIC



Q4 (a)

$$\text{slope ①} = \frac{50 - 0}{20 - 0} = 2.5$$

$$y - y_0 = S_0(x - x_0)$$

$$y - 50 = 2.5(x - 20) \rightarrow \boxed{y = 2.5x}$$

$$\text{slope ②} = \frac{-50 - 50}{60 - 20} = -2.5$$

$$y + 50 = -2.5(x - 60) \rightarrow \boxed{y = 100 - 2.5x}$$

$$\text{slope ③} = \frac{0 + 50}{80 - 60} = 2.5$$

$$y - 0 = 2.5(x - 80) \rightarrow \boxed{y = 2.5x - 200}$$

$$V(t) = \begin{cases} 2.5t & , 0 < t < 20 \\ 100 - 2.5t & , 20 < t < 60 \\ 2.5t - 200 & , 60 < t < 80 \end{cases}$$

$$i(t) = C \frac{dV}{dt}$$

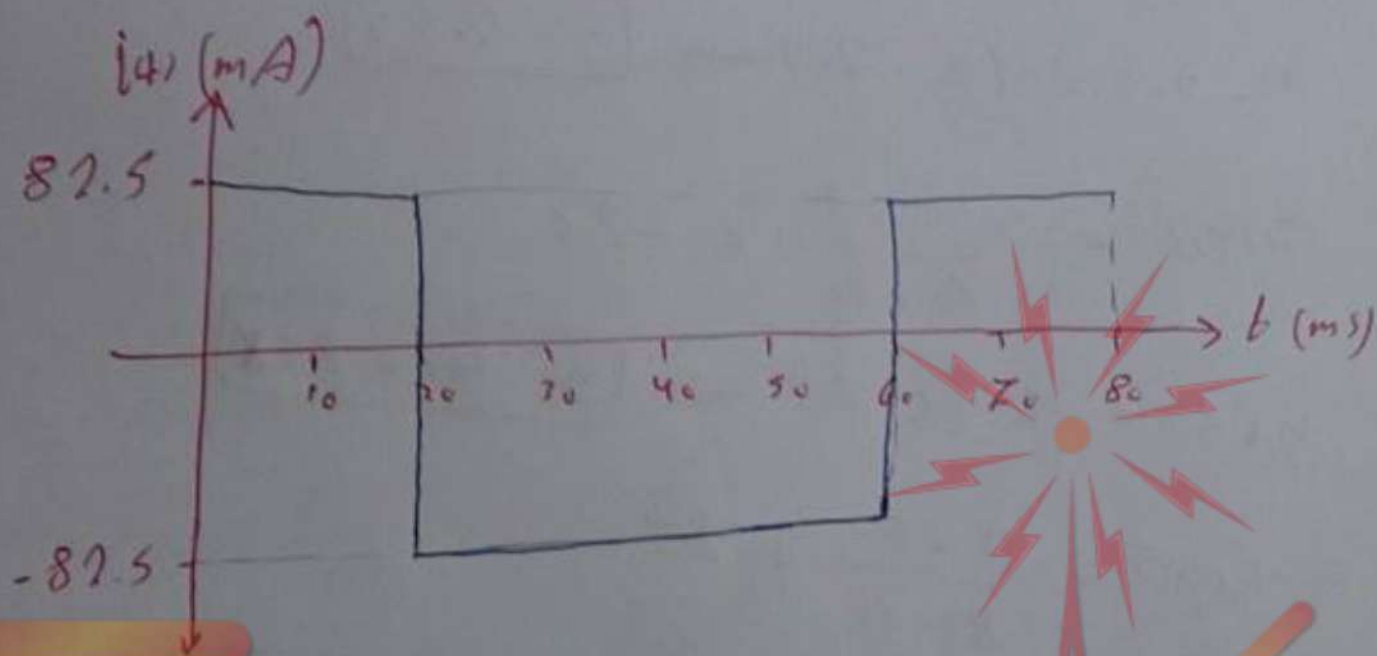
$$i_1(t) = 33\text{m} (2.5) = \boxed{82.5\text{mA}}$$

$$i_2(t) = 33\text{m} (-2.5) = \boxed{-82.5\text{mA}}$$

$$i_3(t) = 33\text{m} (2.5) = \boxed{82.5\text{mA}}$$

Q4

$$i(t) = \begin{cases} 82.5 \text{ mA} & , 0 < t < 20 \\ -82.5 \text{ mA} & , 20 < t < 60 \\ 82.5 \text{ mA} & , 60 < t < 80 \end{cases}$$



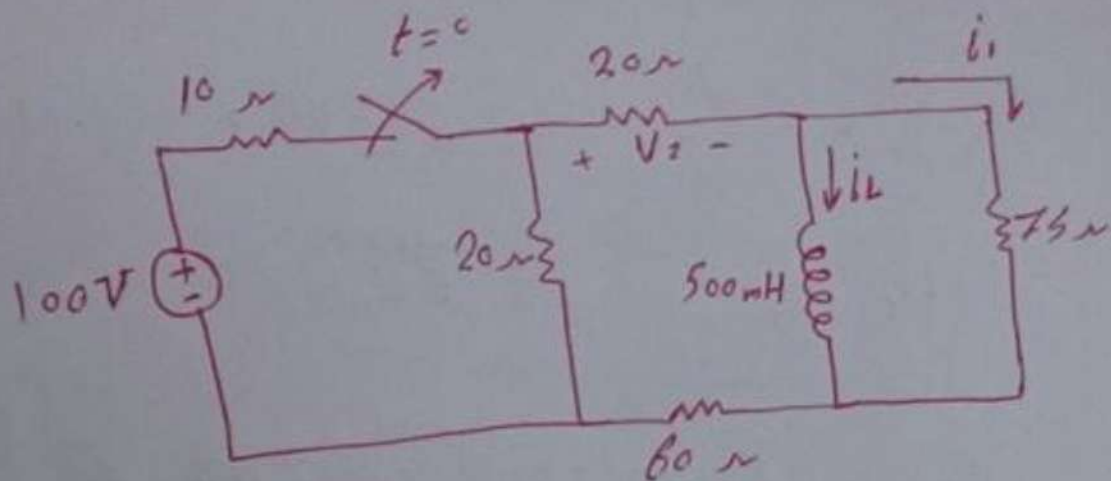
Q4

$$W = \frac{1}{2} CV^2$$

$$W = \frac{1}{2} (33 \text{ m}) (50)^2$$

$$W = 41.25 \text{ J}$$

Q5



$$V_{16\Omega} = \frac{100 \times 16}{16 + 10} = 61.53 \text{ V}$$

$$I_L = \frac{V}{R} = \frac{61.53}{80}$$

$$I_L = 0.769 \text{ A}$$

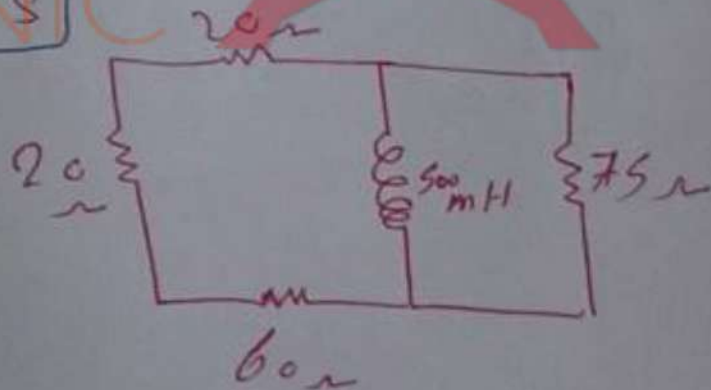
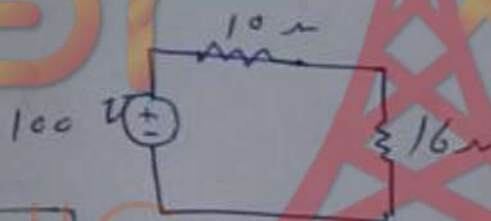
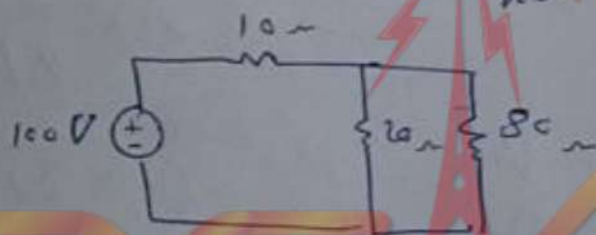
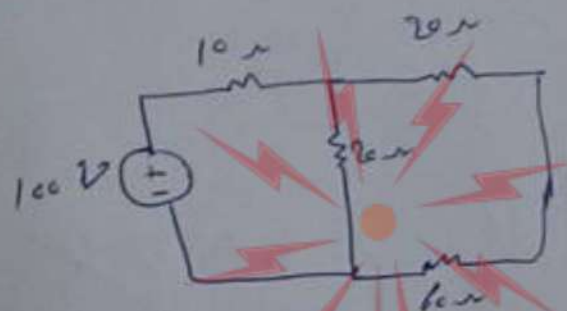
$$R_{th} = 42.85 \Omega$$

$$\tau = \frac{L}{R_{th}} = \frac{500 \text{ mH}}{42.85} = 0.0116 \text{ s}$$

$$I_L = I_{L(0)} e^{-t/\tau}$$

$$I_L = 0.769 e^{-t/0.0116}$$

$$I_L = 0.769 e^{-85.71 t}$$



Q5

$$V_L = L \frac{di}{dt} = 0.5 (0.764 + -85.71 e^{-85.71t})$$

$$V_L = V_{75\Omega} = -33 e^{-85.71t} \text{ Volt}$$

$$I_{75\Omega} = \frac{V}{R} = \frac{-33 e^{-85.71t}}{75} =$$

$$-0.44 e^{-85.71t} \text{ A}$$

$$V_L = V_{100\Omega} =$$

100Ω



75Ω

ولتاژ

$$-33 e^{-85.71t} \text{ Volt}$$

$$I_{100\Omega} = \frac{V}{R} = \frac{-33 e^{-85.71t}}{100} = -0.33 e^{-85.71t} \text{ A}$$

$$V_{20\Omega} = I_{100\Omega} R = -0.33 e^{-85.71t}$$

20Ω

$$V_{20\Omega} = -6.6 e^{-85.71t} \text{ Volt}$$



Electrical circuits 1.

Final Exam
First semester
2016-2017.

Date: Jan. 7, 2016

Exam Time: 2 hours

Choose the correct answer.

* For the questions 1,2,3,4 consider the circuit shown in figure 1 and answer:

1) The value of V_0 is:

- a) 20 V b) 10 V
c) 40 V d) -40 V

2) The value of I is:

- a) 4 A b) 1 A
c) 2 A d) -4 A

3) The power absorbed by 5Ω resistor is:

- a) 100 W b) 40 W c) 160 W

4) The power delivered by 40V voltage source is:

- a) 180 W b) 40 W c) 80 W d) 100 W

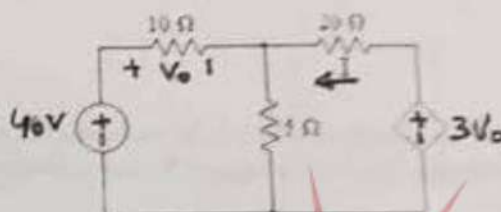


Fig.1

* For the questions 5,6,7,8 consider the circuit shown in figure 2 and answer:

5) The value of R_{TH} with respect to terminals (a,b) is:

- a) 6Ω b) 3Ω
c) 14Ω d) 16Ω

6) The value of V_{TH} from the terminals (a,b) is:

- a) 28 V b) 12 V
c) 32 V d) 6 V

7) The maximum power can be delivered to any load resistor is:

- a) 3 W b) 9 W c) 36 W d) 18 W

8) The current I is:

- a) 3 A b) 1.5 A c) 4.5 A d) 9 A



Fig.2

* For the questions 9,10,11,12 consider the circuit shown in figure 3 and answer:

9) The value of $v(t)$ is:

- a) 20 V b) 24 V
c) 8 V d) 16 V

10) the value of $v(t = \infty)$ is:

- a) 8 V b) 0 V
c) 24 V d) 12 V

11) The value of τ (time constant) is:

- a) 0.5 s b) 0.5 ms
c) 2 ms d) 10 ms

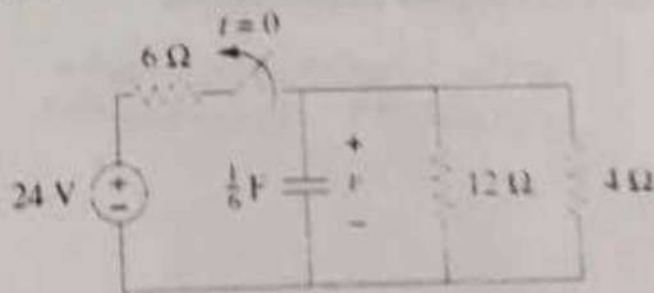


Fig.3

- 12) The value of v at $t = 750\text{ms}$ is: .
 a) 1.41V b) 0.8925 V c) 1.785V d) 3.57V

- 13) Refer to figure 4; the value of the equivalent capacitance is:
 a) $80\mu\text{F}$ b) $40\mu\text{F}$ c) $320\mu\text{F}$ d) $170\mu\text{F}$

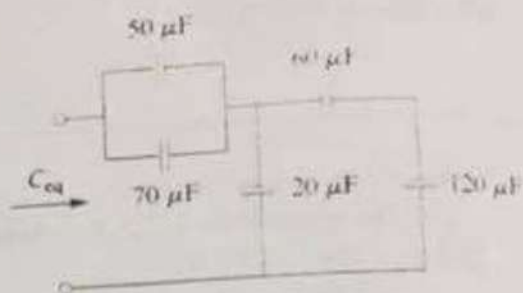


Fig 4

* For the questions 14,15,16 consider the circuit(the circuit under dc steady state conditions) shown in figure 5 and answer:

- 14) The inductor current i_L is:

a) 2.5A b) 10A
 c) 5A d) 7.5A

- 15) The value of the capacitor voltage V_C is:

a) 10V b) 20V
 c) 15V d) 5V

- 16) The energy stored in the capacitor is:

a) 50 J b) 450 J c) 200 J d) 800 J

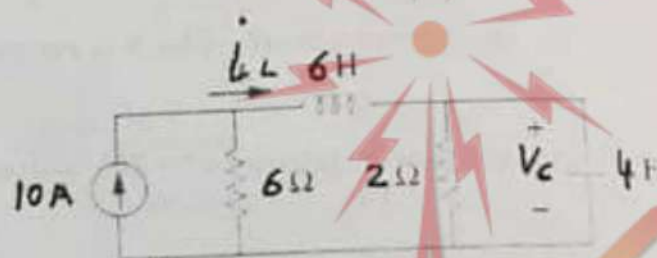


Fig.5

For the questions 17,18,19,20 consider the circuit shown in figure 6. The switch was open for a long time, but closed at $t=0$. If $i(0)=10\text{A}$. Answer:

- 17) The value of $i(t=\infty)$ is:

a) 6A b) 12A
 c) 1A d) 5A

- 18) The time constant is:

a) 500ms b) 400ms
 c) 200ms d) 20 ms

- 19) The value of $i(t=100\text{ms})$ is:

a) 3A b) 6A
 c) 4.2A d) 8.4A

- 20) The complete response $i(t)$ is:

a) $i(t)=12+6e^{-50t}\text{A}$ b) $i(t)=6+4e^{-5t}\text{A}$
 c) $i(t)=6+e^{-5t}\text{A}$ d) $i(t)=6+4e^{-0.5t}\text{A}$

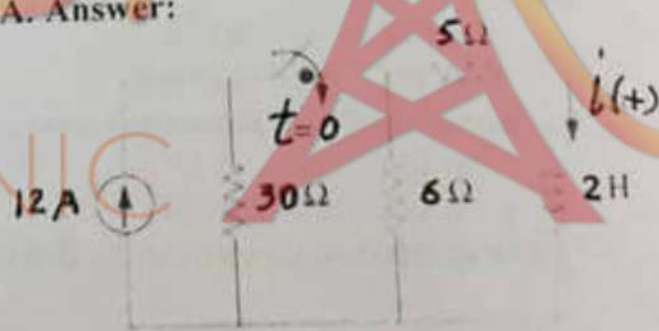


Fig.6

* For the questions 21,22,23,24,25 consider the circuit shown in figure 7. Assuming $v(0)=5V, i(0)=0, L=1H, C=10mF, R=1.923 \Omega$. Answer:

21) The value of α (damping factor) is:

- a) 52 b) 6.5
c) 13 d) 26

22) The value of ω_o (resonant frequency) is:

- a) 40 b) 10
c) 20 d) 5

23) The value of S1 is:

- a) -6 b) -4 c) -2 d) -25

24) The value of S2 is:

- a) -100 b) -25 c) -12.5 d) -50

25) The natural response $v(t)$ for $t > 0$ is:

- a) $v(t) = -0.2083e^{-2t} + 5.208e^{-50t}$ b) $v(t) = -0.2083e^{-4t} + 5.208e^{-25t}$
c) $v(t) = 0.2083e^{-2t} - 5.208e^{-50t}$ d) $v(t) = 0.2083e^{-4t} - 5.208e^{-25t}$

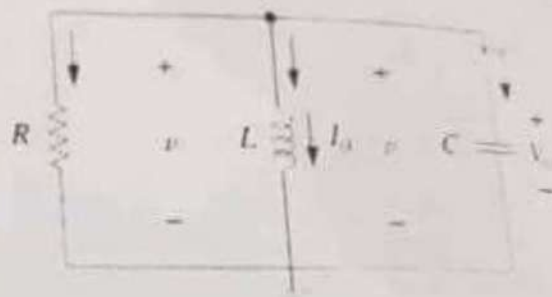


Fig.7

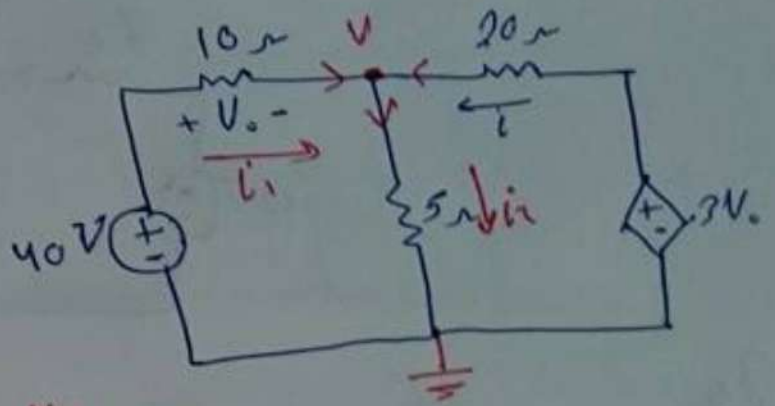
1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	

Good Luck

POLYTECHNIC

Q1

$$\frac{40 - V}{10} + \frac{3(40 - V) - V}{20} = \frac{V}{5}$$



$$80 - 2V + 120 - 3V - V - 4V = 0$$

$$10V = 200 \rightarrow V = 20 \text{ Volt}$$

$$V_0 = 40 - V$$

$$V_0 = 40 - 20 = 20 \text{ Volt}$$

Q2

$$I = \frac{3V_0 - V}{20} = \frac{3 \times 20 - 20}{20} = 2 \text{ A}$$

Q3

$$i_1 = \frac{40 - 20}{10} = 2 \text{ A}$$

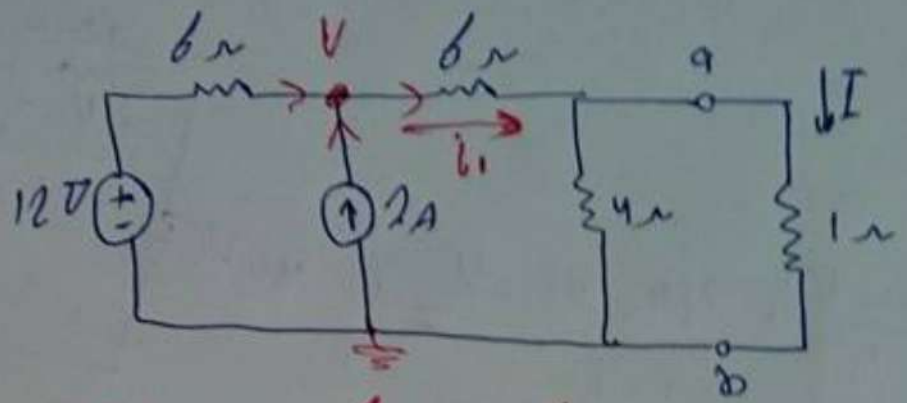
$$i_2 = i_1 + I = 2 + 2 = 4 \text{ A}$$

$$P = I^2 R = (4)^2 \times 5 = 80 \text{ watt}$$

Q4

$$P = VI = -40 \times 2 = -80 \text{ watt}$$

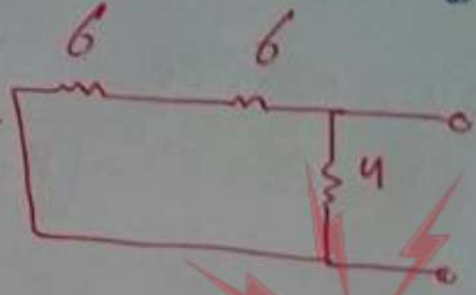
$$P_{\text{delivered}} = -(-80) = 80 \text{ watt}$$



Q5

$$R_{th} = [6\Omega \text{ series } 6\Omega] \parallel 4\Omega$$

$$R_{th} = 3\Omega$$



Q6

$$\frac{12 - V}{6} + 2 = \frac{V}{4} \rightarrow 8V = 120$$

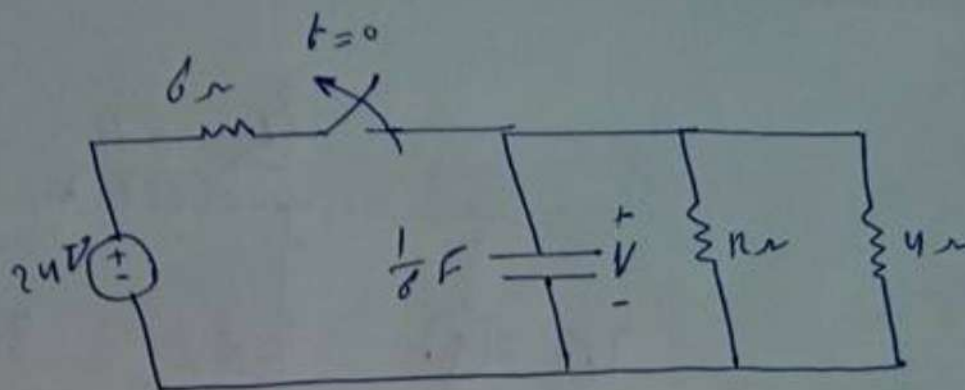
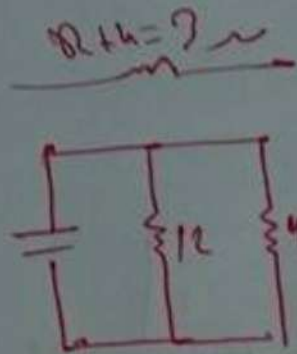
$$V = 15 \text{ Volt}$$

$$I_1 = \frac{V}{R} = \frac{15}{10} = 1.5 \text{ A}$$

$$V_{th} = V_{4\Omega} = 1.5 \times 4 = 6 \text{ Volt}$$

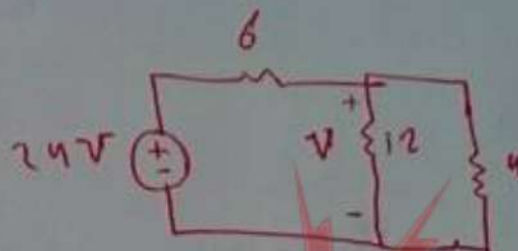
$$P_{max.} = \frac{V_{th}^2}{4 R_{th}} = \frac{6^2}{4(3)} = 3 \text{ Watt}$$

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{6}{3 + 1} = \frac{6}{4} = 1.5 \text{ A}$$



Q9

$$V_0 = \frac{V \times R_{eq}}{R_{eq} + R} = \frac{24 \times 3}{6 + 3} = \boxed{8 \text{ Volt}}$$



Q10 $V = V_0 e^{-t/\tau} = 8 e^{-\infty/0.5}$

$$= 8 e^{-\infty} = \frac{8}{\infty} = \boxed{0 \text{ Volt}}$$

Q11 $\tau = 0.5 \text{ s}$

Q12 $V = V_0 e^{-t/\tau} = 8 e^{-\frac{750 \times 10^{-3}}{0.5}} = \boxed{1.785 \text{ Volt}}$

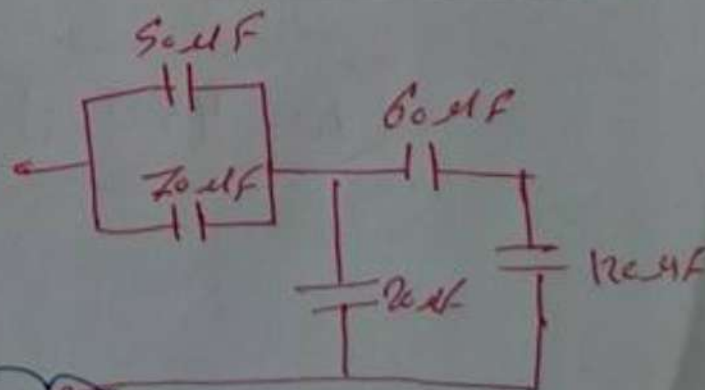
Q13

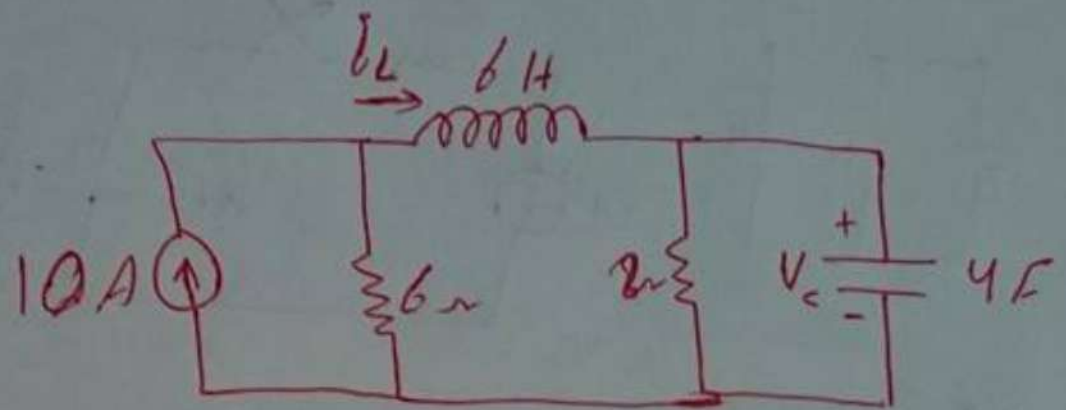
$[60\mu \text{ series } 120\mu] = 40\mu\text{F}$

$[40\mu \parallel 20\mu] = 60\mu\text{F}$

$[50\mu \parallel 70\mu] = 120\mu\text{F}$

$[120\mu \text{ series } 60\mu] = \boxed{40\mu\text{F}}$

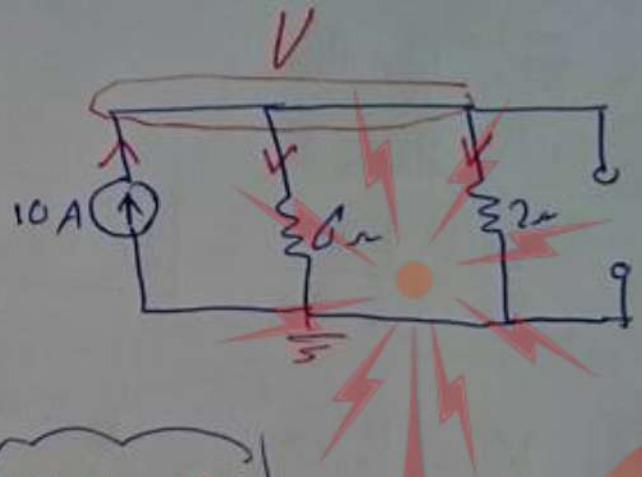




Q14

$$10 = \frac{V}{6} + \frac{V}{2}$$

$$\rightarrow V = 15 \text{ Volt}$$



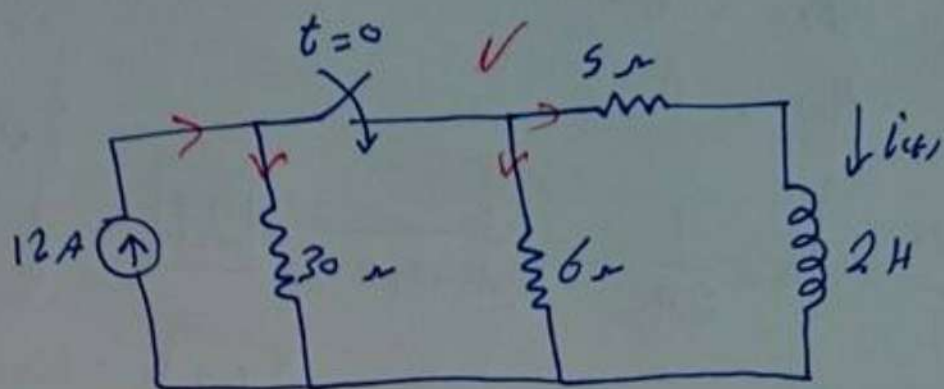
$$i_L = i_{2\Omega} = \frac{V}{R} = \frac{15}{2} = 7.5 \text{ A}$$

Q15 $V_c = V = 15 \text{ Volt}$

Q16 $W = \frac{1}{2} C V^2 = \frac{1}{2} \times 4 \times (15)^2$

$W = 450 \text{ J}$

Q17



$$12 = \frac{V}{30} + \frac{V}{6} + \frac{V}{5} \rightarrow V = 30 \text{ Volts}$$

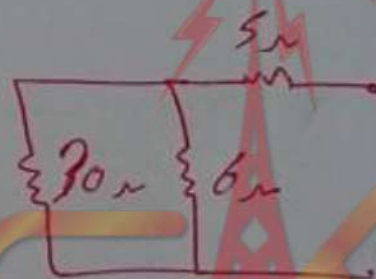
$$i_L(t) = \frac{V}{R} = \frac{30}{5} = 6 \text{ A}$$

$$i(\infty) = i_L(t) = 6 \text{ A}$$

Q18

$R_{th} = (30 \parallel 6) \text{ series } 5$

$$R_{th} = 10 \Omega$$



$$\tau = \frac{L}{R_{th}} = \frac{2}{10} = 0.2 = 200 \text{ ms}$$

Q19

$$i(100 \text{ ms}) = i(\infty) + (i_0 - i(\infty)) e^{-t/\tau} = 6 + (10 - 6) e^{-\frac{100 \text{ ms}}{0.2}} = 8.4 \text{ A}$$

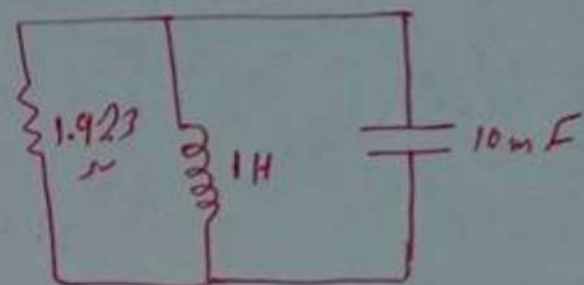
Q20

$$i_L(t) = i(\infty) + (i_0 - i(\infty)) e^{-t/\tau}$$

$$i_L(t) = 6 + 4 e^{-5t}$$

$$V(0) = 5 \text{ Vdc}$$

$$i(0) = 0$$



$$Q_{21} \quad \alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times 10^{-3}} = \boxed{26}$$

$$Q_{22} \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10^{-3}}} = \boxed{10}$$

$$Q_{23} \quad s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -26 + \sqrt{26^2 - 10^2} = \boxed{-2}$$

$$Q_{24} \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -26 - \sqrt{26^2 - 10^2} = \boxed{-50}$$

$$Q_{25} \quad \alpha > \omega \rightarrow \text{over damped}$$

$$V(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$V(t) = A_1 e^{-2t} + A_2 e^{-50t}$$

$$V(0) = A_1 + A_2 \rightarrow \boxed{5 = A_1 + A_2}$$

$$\frac{dV}{dt} = -2A_1 e^{-2t} - 50A_2 e^{-50t}$$

موضات في الكهارة

$$\frac{dV(0)}{dt} = -2A_1 - 50A_2$$

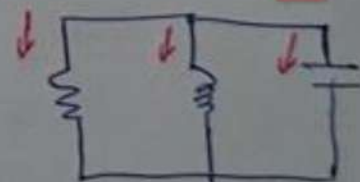
$$\boxed{0 = -2A_1 - 50A_2} \quad \text{--- (2)}$$

$$A_1 = 5.2$$

$$A_2 = -0.2$$

$$V(t) = 5.2 e^{-2t} - 0.2 e^{-50t}$$

الاجابة النهائية



$$0 = i_R + i_L + i_C$$

$$0 = \frac{V}{1.923} + 0 + 10^{-3} \frac{dV}{dt}$$

موضات في الكهارة

$$\boxed{\frac{dV(0)}{dt} = 0}$$



Q1: Find the voltage (v) for the network shown in Fig 1 using superposition theorem. (10marks)

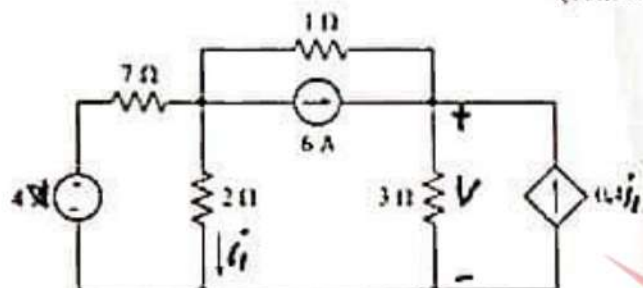


Fig 1

Q2: For the circuit shown in Fig 2:
a) Determine the value of R_1 to which a maximum power can be delivered.
b) The maximum power delivered to R_1 . (10marks)

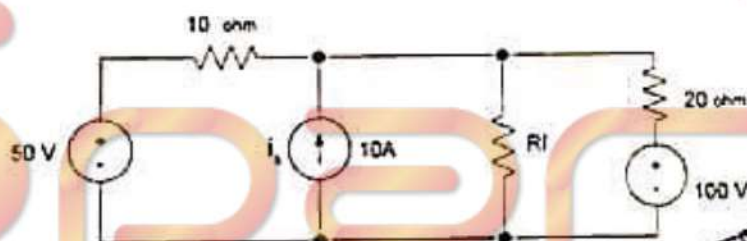


Fig 2

Q3: a) For the circuit shown in Fig 3.a find the equivalent capacitance (10 marks)

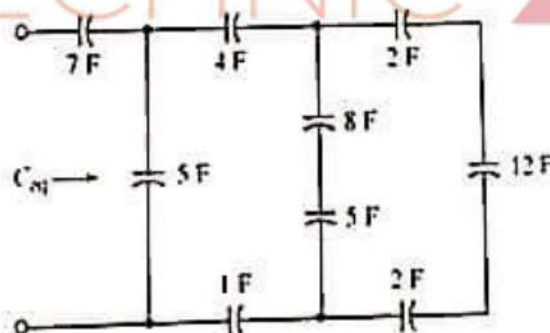


Fig 3.a

b) Calculate the power dissipated in the $40\ \Omega$ resistor and the voltage labeled V_C for the circuit given in fig 3.b

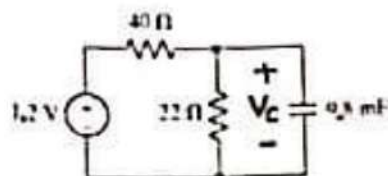


fig 3 b

Q4: The voltage across a $33\ \text{mF}$ capacitor is shown graphically in Fig 4: (8 marks)

a) Assuming the passive sign convention, sketch the resulting current waveform across the capacitor.

b) Find the energy stored in the capacitor at $t=20\text{ms}$.

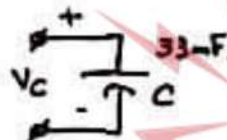
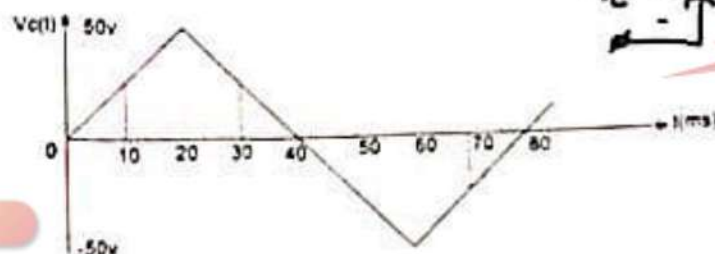


Fig 4

Q5: For the circuit shown in Fig 5 the switch opens at $t=0$ after having been closed for a long time. Find: (12 marks)

- $i_1(t)$ for $t \geq 0$
- $i_L(t)$ for $t \geq 0^+$
- $V_2(t)$ for $t \geq 0^+$

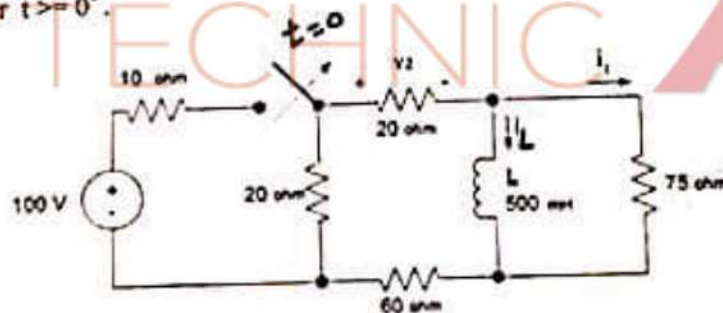


Fig 5

Good Luck

Q11

[افغان فانیج
Circuit 1]

Sultani

4/8/2015

$$U = U' + U''$$

$$= 0.69 + 2.682$$

$$U = 3.372 \text{ V}$$

node (U_1)

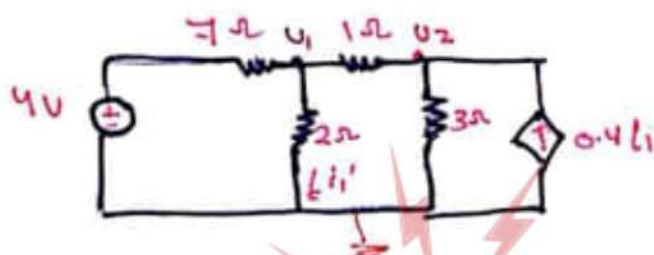
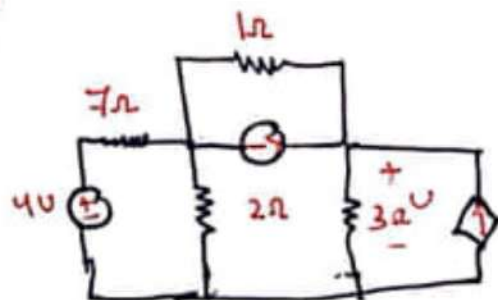
$$\frac{4 - U_1}{7} = \frac{U_1}{2} + \frac{U_1 - U_2}{1}$$

$$23U_1 - 14U_2 = 8$$

node (U_2)

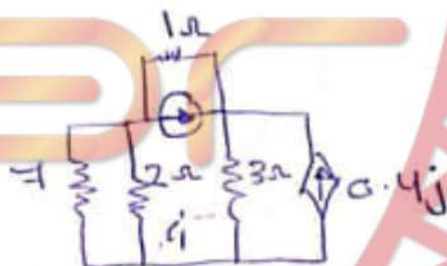
$$U_1 - U_2 + 0.4 \frac{U_1}{2} = \frac{U_2}{3} \rightarrow 7.2U_1 - 8U_2 = 0$$

$$U_1 = 0.76 \text{ V} \quad | \quad U' = U_2 = 0.69 \text{ volt}$$

Sultani abushob
#spark-team

→ Mesh ①

$$I_1 = 0.4 \text{ A} = 0.4 (I_2 - I_3)$$



→ Mesh (2)

$$8I_2 - 2I_3 - 3I_1 = -6$$

$$8I_2 - 2I_3 - 3(0.4(I_2 - I_3)) = -6$$

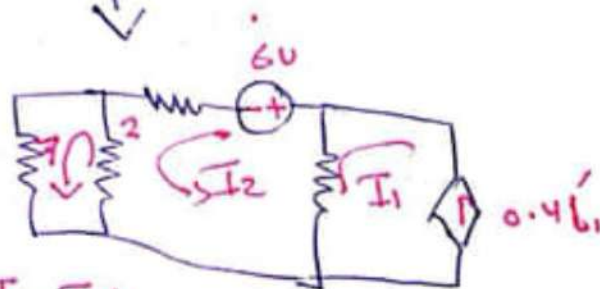
$$4.8I_2 - 0.8I_3 = -6$$

→ Mesh ③

$$-2I_2 + 9I_1 = 0$$

$$I_2 = -1.298 \text{ A}$$

$$I_3 = -0.288 \text{ A}$$



$$U'' = 3(I_1 - I_2)$$

$$U'' = 3(-0.404 + 1.298)$$

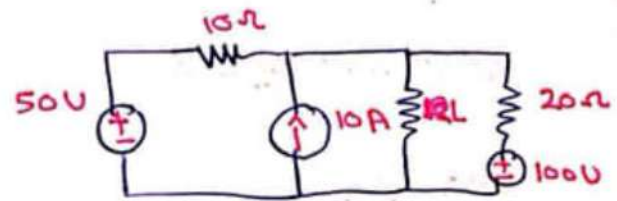
$$U'' = 2.682 \text{ V}$$

$$I_1 = 0.4(-1.298 + 0.288)$$

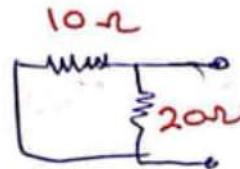
$$I_1 = -0.404$$

①

Q.2

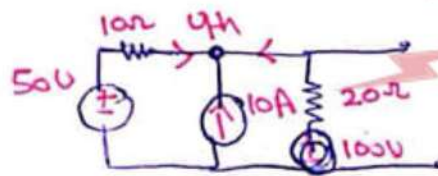


$$R_{th} = R_L = 6.67 \Omega$$



node (V_{th})

$$\frac{50 - V_{th}}{10} + 10 + \frac{100 - V_{th}}{20} = 0$$



$$100 - 2V_{th} + 200 + 100 - V_{th} = 0$$

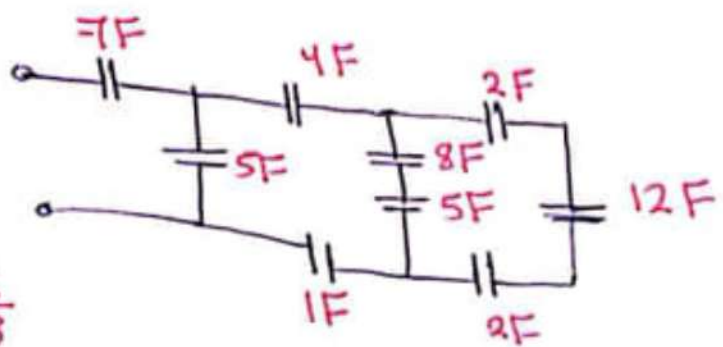
$$3V_{th} = 400 \rightarrow V_{th} = 133.34 \text{ Volt}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{(133.34)^2}{4 \times 6.67}$$

$$P_{max} = 666.7 \text{ watt}$$

Sultan
abushanab

Q3 a)



$$[2 \text{ series } 12 \text{ series } 2] \rightarrow \frac{12}{13}$$

$$[8 \text{ series } 5] \rightarrow \frac{40}{13}$$

$$\left[\frac{12}{13} \parallel \frac{40}{13} \right] \rightarrow 4$$

$$[4 \text{ series } 4 \text{ series } 1] \rightarrow \frac{2}{3}$$

$$\left[\frac{2}{3} \parallel 5 \right] \rightarrow \frac{17}{3}$$

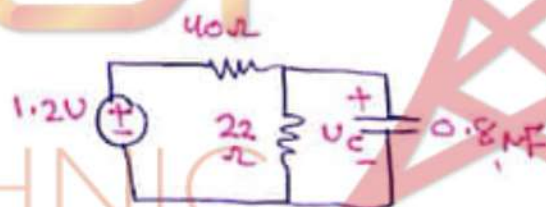
$$C_{eq} = \left[\frac{17}{3} \text{ series } 7 \right]$$

$$C_{eq} = 3.13 \text{ F}$$



Q3 b)

$$U_{40\Omega} = \frac{1.2 \times 40}{40 + 22} = \frac{24}{31} \text{ Volt}$$



$$P_{uo} = \frac{U^2}{R} = 0.0149 \text{ Watt}$$

$$U_C = U_{22\Omega} = \frac{1.2 \times 22}{22 + 40} = 0.425 \text{ Volt}$$

Q4] a

step ①

$$= \frac{50 - 0}{20 - 0} = 2.5$$

$$y - y_0 = S_c (x - x_0) \rightarrow y - 50 = 2.5 (x - 20) \dots \rightarrow y = 2.5x$$

step ②

$$= \frac{-50 - 50}{60 - 20} = -2.5$$

$$y + 50 = -2.5 (x - 60) \rightarrow \boxed{y = 100 - 2.5x}$$

step 3

$$= \frac{0 + 50}{80 - 60} = 2.5$$

$$y - 0 = 2.5 (x - 80) \rightarrow \boxed{y = 2.5x - 200}$$

$$v(t) = \begin{cases} 2.5t & , 0 < t < 20 \\ 100 - 2.5t & , 20 < t < 60 \\ 2.5t - 200 & , 60 < t < 80 \end{cases}$$

$$i(t) = C \frac{dv}{dt}$$

$$i_1(t) = 33m(2.5) = 82.5mA$$

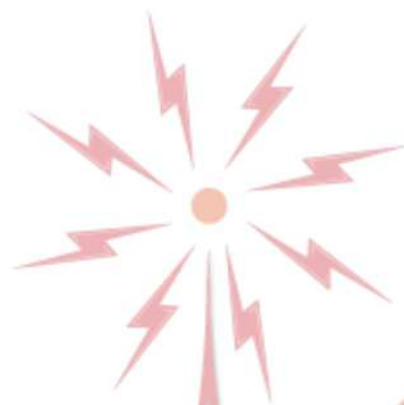
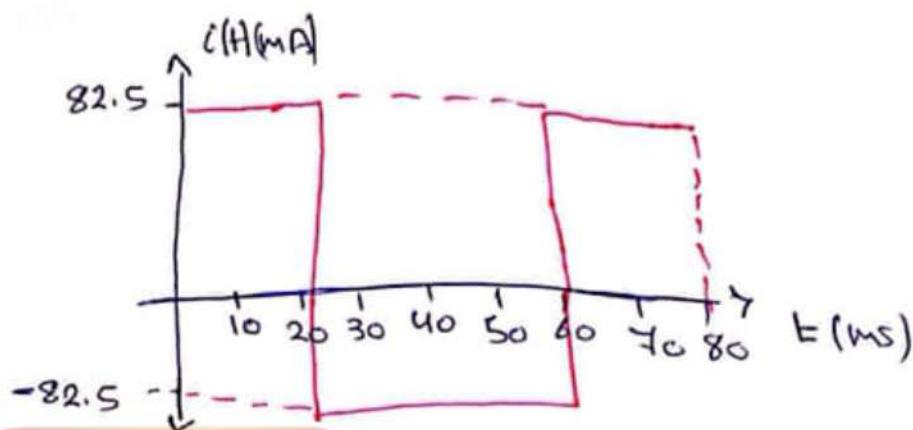
$$i_2(t) = 33m(-2.5) = -82.5mA$$

$$i_3(t) = 33m(2.5) = 82.5mA$$

④

Q4)

$$i(t) = \begin{cases} 82.5 \text{ mA} & , 0 < t < 20 \\ -82.5 \text{ mA} & , 20 < t < 60 \\ 82.5 \text{ mA} & , 60 < t < 80 \end{cases}$$



⑥

$w = \frac{1}{2} C U^2$
POLYTECHNIC

$$w = \frac{1}{2} (33 \text{ m}) (50)^2 = 41.25 \text{ J}$$

Sul Karabustan b

⑤

Q5

$$U_{16\Omega} = \frac{100 \times 16}{16+10} = 61.53 \text{ V}$$

$$I_{L_0} = I_{8\Omega} = \frac{U}{R} = \frac{61.53}{80}$$

$$I_{L_0} = 0.769 \text{ A}$$

$$R_{th} = 42.85 \Omega$$

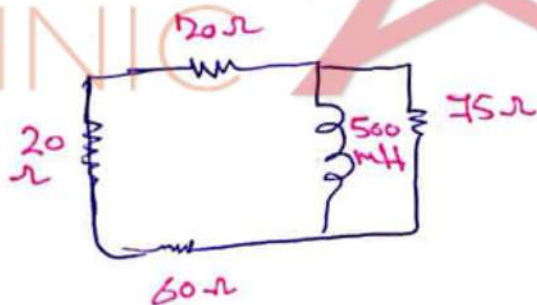
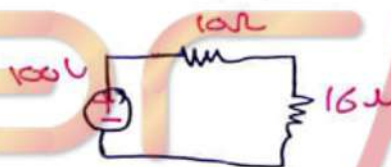
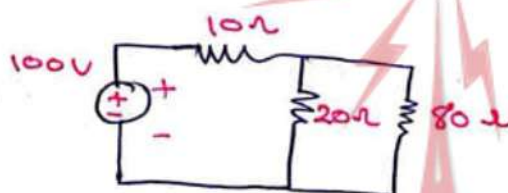
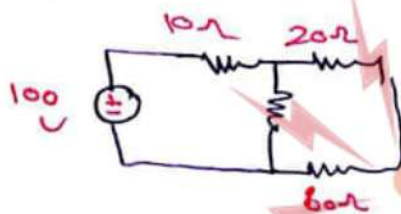
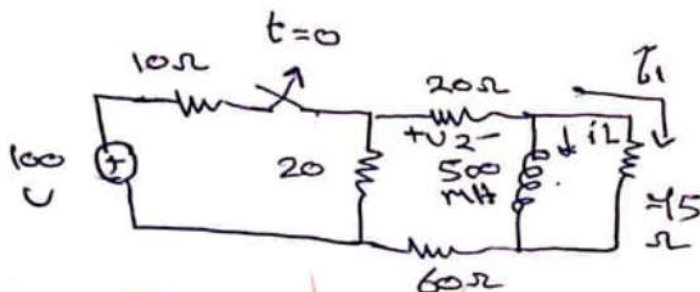
$$\tau = \frac{L}{R_{th}} = \frac{500 \text{ mH}}{42.85}$$

$$= 0.01185$$

$$I_L = I_L(0) e^{-t/\tau}$$

$$I_L = 0.769 e^{-t/0.01185}$$

$$I_L = 0.769 e^{-85.71 t}$$



②

DATE, TIME
FAX NO. / NAME
DURATION
PAGE(S)
RESULT
MODE

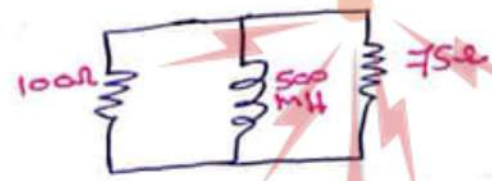
29/10 14:24
414
00:00:21
82
OK
STANDARD
ECM

P5

$$U_L = L \frac{di}{dt} = 0.5 (0.769 + -85.71 e^{-85.71t})$$

$$U_L = U_{75\Omega} = -\frac{33 e^{-85.71t}}{75} = -0.44 e^{-85.71t} \text{ A}$$

$U_L = U_{100\Omega}$
 $= -33 e^{-85.71t} \text{ Volt}$



$$I_{100\Omega} = \frac{U}{R} = \frac{-33 e^{-85.71t}}{100} = -0.33 e^{-85.71t} \text{ A}$$

$$U_{20\Omega} = I_{100\Omega} R = -0.33 e^{-85.71t} \times 20$$

$$U_{20\Omega} = -6.6 e^{-85.71t} \text{ Volt}$$

7



Choose the correct answer.

* For the questions 1,2,3,4 consider the circuit shown in figure 1 and answer:

1) The value of V_o is:

- a) 20 V b) 10 V
c) 40 V d) -40 V

2) The value of I is:

- a) 4A b) 1A
c) 2A d) -4A

3) The power absorbed by 5Ω resistor is:

- a) 100W b) 40W c) 160W

4) The power delivered by 40V voltage source is:

- a) 180W b) 40W c) 80W d) 100W

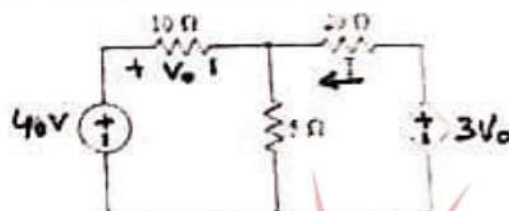


Fig.1

* For the questions 5,6,7,8 consider the circuit shown in figure 2 and answer:

5) The value of R_{TH} with respect to terminals (a,b) is:

- a) 6Ω b) 3Ω
c) 14Ω d) 16Ω

6) The value of V_{TH} from the terminals (a,b) is:

- a) 28V b) 12V
c) 32V d) 6V

7) The maximum power can be delivered to any load resistor is:

- a) 3W b) 9W c) 36W d) 18W

8) The current I is:

- a) 3A b) 1.5A c) 4.5A d) 9A

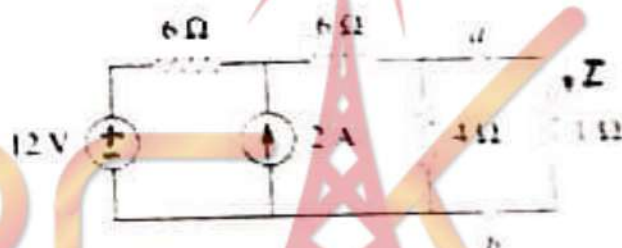


Fig.2

* For the questions 9,10,11,12 consider the circuit shown in figure 3 and answer:

9) The value of $v(0)$ is:

- a) 20V b) 24V
c) 8V d) 16V

10) the value of $v(t = \infty)$ is:

- a) 8V b) 0V
c) 24V d) 12V

11) The value of τ (time constant) is:

- a) 0.5s b) 0.5 ms
c) 2 ms d) 10 ms

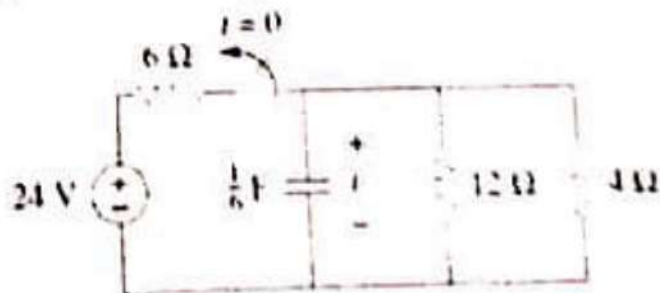


Fig.3

- 12) The value of v at $t = 750\text{ms}$ is:
 a) 1.41V b) 0.8925 V c) 1.785V d) 3.57V

- 13) Refer to figure 4; the value of the equivalent capacitance is:
 a) $80\mu\text{F}$ b) $40\mu\text{F}$ c) $320\mu\text{F}$ d) $170\mu\text{F}$

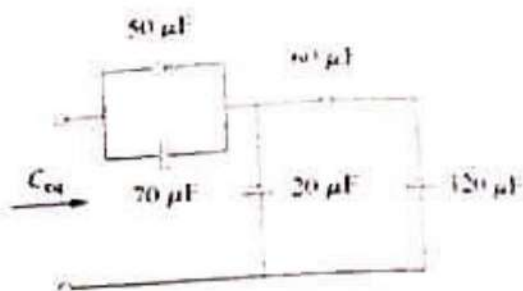


Fig 4

* For the questions 14,15,16 consider the circuit (the circuit under dc steady state conditions) shown in figure 5 and answer:

- 14) The inductor current i_L is:

a) 2.5A b) 10A
 c) 5A d) 7.5A

- 15) The value of the capacitor voltage V_C is:

a) 10V b) 20V
 c) 15V d) 5V

- 16) The energy stored in the capacitor is:

a) 50 J b) 450 J c) 200 J d) 800 J

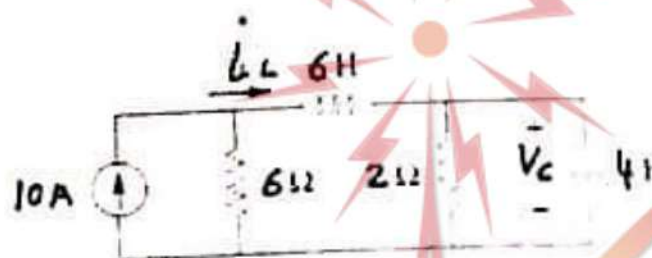


Fig.5

For the questions 17,18,19,20 consider the circuit shown in figure 6. The switch was open for a long time, but closed at $t=0$. If $i(0)=10\text{A}$. Answer:

- 17) The value of $i(t=\infty)$ is:

a) 6A b) 12A
 c) 1A d) 5A

- 18) The time constant is:

a) 500ms b) 400ms
 c) 200ms d) 20 ms

- 19) The value of $i(t=100\text{ms})$ is:

a) 3A b) 6A
 c) 4.2A d) 8.4A

- 20) The complete response $i(t)$ is:

a) $i(t)=12+6e^{-50t}\text{ A}$ b) $i(t)=6+4e^{-5t}\text{ A}$
 c) $i(t)=6+e^{-5t}\text{ A}$ d) $i(t)=6+4e^{-0.5t}\text{ A}$

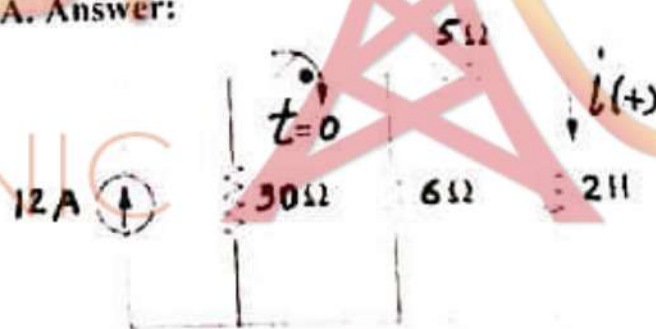


Fig.6

* For the questions 21,22,23,24,25 consider the circuit shown in figure 7. Assuming $v(0)=5V, i(0)=0, L=1H, C=10mF, R=1.923\Omega$. Answer:

21) The value of α (damping factor) is:

- a) 52 b) 6.5
c) 13 d) 26

22) The value of ω_o (resonant frequency) is:

- a) 40 b) 10
c) 20 d) 5

23) The value of S1 is:

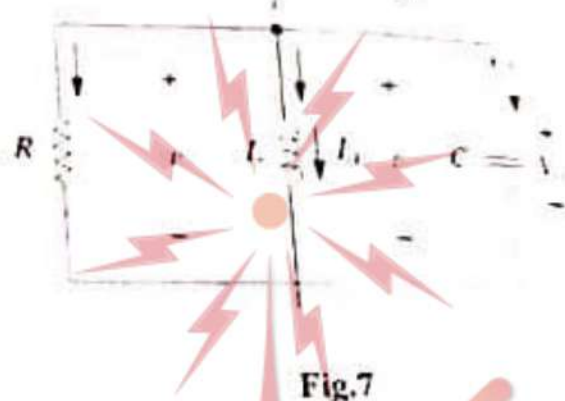
- a) -6 b) -4 c) -2 d) -25

24) The value of S2 is:

- a) -100 b) -25 c) -12.5 d) -50

25) The natural response $v(t)$ for $t > 0$ is:

- a) $v(t) = -0.2083e^{-2t} + 5.208e^{-50t}$ b) $v(t) = -0.2083e^{-4t} + 5.208e^{-25t}$
c) $v(t) = 0.2083e^{-2t} - 5.208e^{-50t}$ d) $v(t) = 0.2083e^{-4t} - 5.208e^{-25t}$



1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	

Good Luck

$$\frac{40 - U}{10} + 3 \frac{(40 - U) - U}{20} = \frac{U}{5}$$

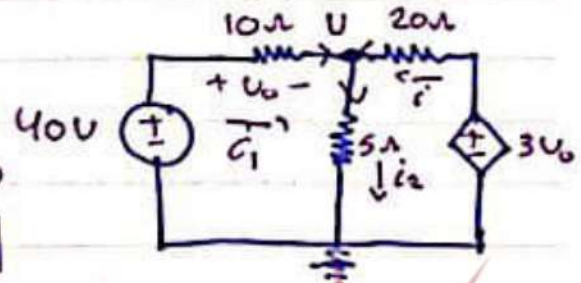
$$80 - 2U + 120 - 3U - U - 4U = 0$$

$$10U = 200 \rightarrow U = 20 \text{ Volt}$$

$$\therefore U_0 = 40 - 20$$

$$= 20 \text{ Volt}$$

11A



Q2 $I = \frac{3U_0 - U}{20} = \frac{3 \times 20 - 20}{20}$

$$= 2 \text{ A}$$

11C

Q3 $i_1 = \frac{40 - 20}{10}$
 $= 2 \text{ A}$

$$i_2 = i_1 + i = 2 + 2 = 4 \text{ A}$$

$$P = I^2 R = (4)^2 \times 5 = 80 \text{ watt}$$

11B

Q4 $P = UI$

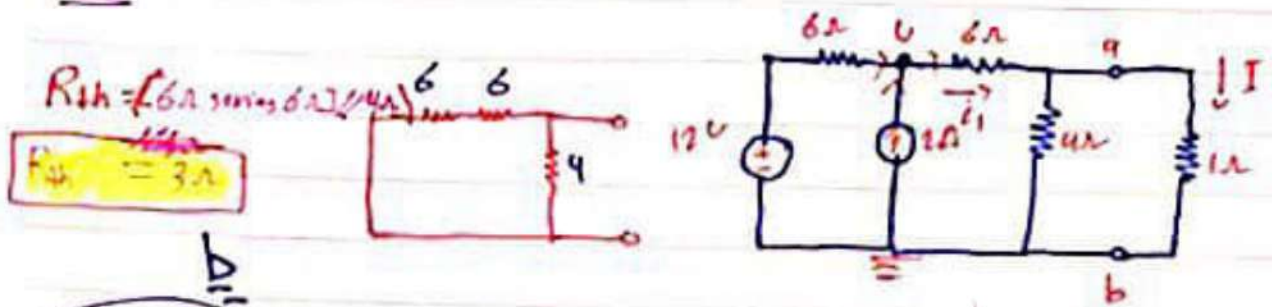
$$= -40 \times 2 = -80 \text{ watt}$$

$$P_{\text{delivered}} = -(-80) = 80 \text{ watt}$$

11

11

Q5



Q6

$$\frac{12 - V}{6} + 12 = \frac{V}{10}$$

$$\rightarrow 8V = 120 \therefore V = 15 \text{ Volt}$$

$$I_1 = \frac{V}{R} = \frac{15}{10} = 1.5 \text{ A}$$

$$\rightarrow V_{th} = V_{4\Omega} = 1.5 \times 4 = 6 \text{ Volt}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{6^2}{4(3)} = 3 \text{ Watt}$$

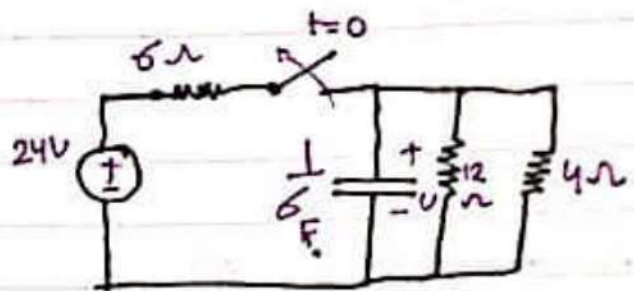
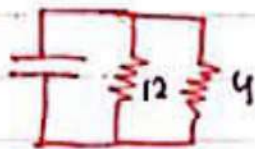
Q8

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{6}{3 + 1} = \frac{6}{4} = 1.5 \text{ A}$$

Sulhan



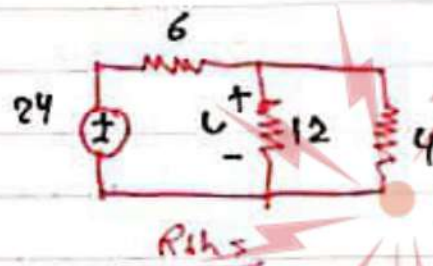
* $R_{th} = 3 \Omega$



Q9

$$U_0 = U \times \frac{R_{eq}}{R_{eq} + R}$$

$$= \frac{24 \times 3}{6 + 3} = 8 \text{ Volt}$$



Q10

$$U = U_0 \cdot e^{-t/\tau} = 8e^{-\infty/0.5}$$

$$= 8e^{-\infty} = \frac{8}{\infty} = 0 \text{ Volt}$$

$$\tau = R_{th} \cdot C$$

$$= 3 \times \frac{1}{6}$$

$$= 0.5 \text{ s}$$

Q11

$$\tau = 0.5 \text{ s}$$

Q12

$$U = U_0 \cdot e^{-t/\tau} = 8e^{-\frac{750 \times 10^{-3}}{0.5}}$$

$$= 1.785 \text{ Volt}$$

$$\left(\frac{1}{2} C V^2 = 45 \right)$$

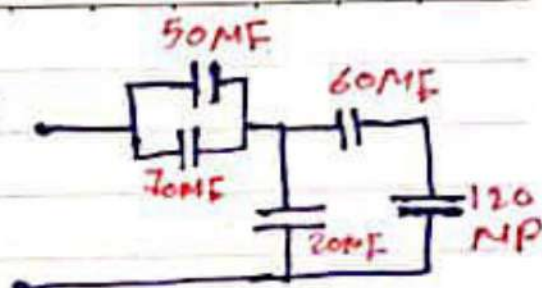
Q13)

$$[60\text{MF series } 120\text{MF}] = 40\text{MF}$$

$$[40\text{MF} // 20\text{MF}] = 60\text{MF}$$

$$[50\text{MF} // 70\text{MF}] = 120\text{MF}$$

$$[120\text{MF series } 60\text{MF}] = 40\text{MF}$$

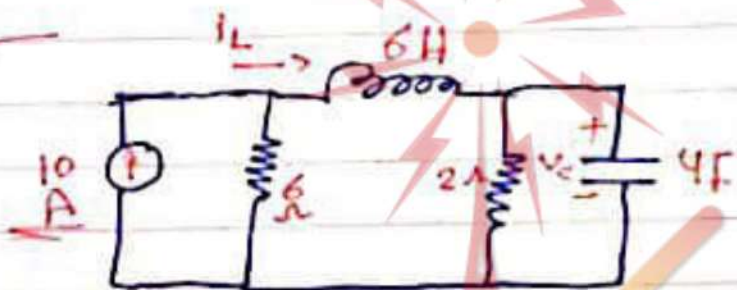


Q14

$$10 = \frac{U}{6} + \frac{U}{2}$$

$$\therefore U = 15 \text{ Volt}$$

$$I_L = I_{2\Omega} = \frac{U}{2} = \frac{15}{2} = 7.5 \text{ A}$$

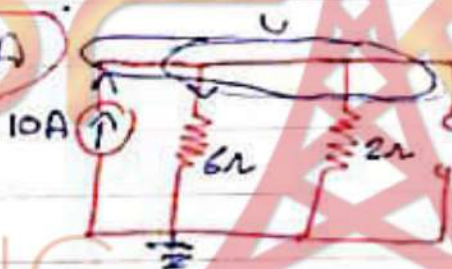


Q15)

$$U_C = U = 15 \text{ Volt}$$

(التوازي)

C



Q16

$$W = \frac{1}{2} C V^2 = \frac{1}{2} \times 4 (15)^2$$

$$= W = 450 \text{ J}$$

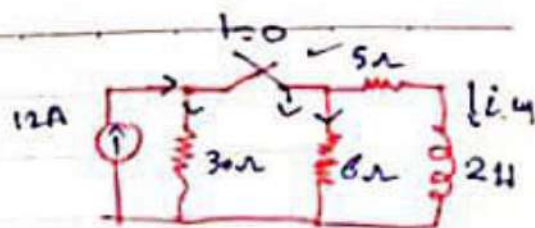
b

1. Current flow

Q17)

$$12 = \frac{V}{30} + \frac{V}{6} + \frac{V}{5}$$

$$\therefore V = 30 \text{ Volt}$$



$$i(t) = \frac{V}{R} = \frac{30}{5} = 6 \text{ A}$$

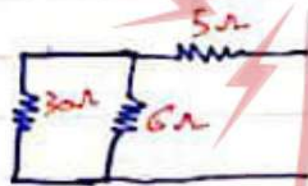
$$i(\infty) = i(0) = 6 \text{ A}$$

Q18

$R_{th} = (30 // 6)$ series

$$R_{th} = 10 \Omega$$

$$\tau = \frac{L}{R_{th}} = \frac{2}{10} = 0.2 = 200 \text{ ms}$$



$$Q19) i(100 \text{ ms}) = i(\infty) + (i_0 - i(\infty))e^{-t/\tau}$$

$$= 6 + (10 - 6)e^{-\frac{100 \text{ ms}}{0.2}} = 8.4 \text{ A}$$

d

Q20

$$i(t) = i(\infty) + (i_0 - i(\infty))e^{-t/\tau}$$

$$i(t) = 6 + 4e^{-5t}$$

b

(5)

No

فائل: سید

$$V(b) = 5V, R = 1.923 \Omega$$

$$I(0) = 0, L = 1H$$

$$V(0) = 5V$$



$$Q_{21} \quad \alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times 10^{-6}} = \boxed{26} \quad d$$

$$Q_{22} \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10^{-6}}} = \boxed{10} \quad b$$

$$Q_{23} \quad S_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -26 + \sqrt{26^2 - 10^2} = \boxed{-2} \quad c$$

$$Q_{24} \quad S_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -26 - \sqrt{26^2 - 10^2} = \boxed{-50} \quad d$$

No

فائل سہیت =

$$V(b) = 5V, R = 1.923 \Omega$$

$$I(0) = 0, L = 1H$$

$$V(0) = 5V$$



$$Q_{21} \quad \alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times 10^{-6}} = \boxed{26} \quad d$$

$$Q_{22} \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10^{-6}}} = \boxed{10} \quad b$$

$$Q_{23} \quad S_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -26 + \sqrt{26^2 - 10^2} = \boxed{-2} \quad c$$

$$Q_{24} \quad S_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -26 - \sqrt{26^2 - 10^2} = \boxed{-50} \quad d$$

Q25) $\alpha > \omega \rightarrow$ over damped

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$v(t) = A_1 e^{-2t} + A_2 e^{-50t}$$

$$v(0) = A_1 + A_2 \rightarrow 5 = A_1 + A_2 \quad (1)$$

$$\frac{dv}{dt} = -2A_1 e^{-2t} - 50A_2 e^{-50t}$$

$$\frac{dv(0)}{dt} = \frac{-v_0 - R i_0}{R C}$$

$$= \frac{-5 - R \times 0}{1.923 \times 10 \times 10^{-3}} = -260$$

$$-260 \cdot 0.1 = 2A_1 - 50A_2$$

$$5 = A_1 + A_2$$

$$\rightarrow v = 5.2 e^{-2t} + 0.2 e^{-50t}$$

$$\frac{dv(0)}{dt} = -2A_1 - 50A_2$$

$$-260 = -2A_1 - 50A_2$$

$$(2)$$

Q1. Use superposition theorem to find i_0 and the power absorbed by the 6Ω resistor in the circuit shown in figure 1.

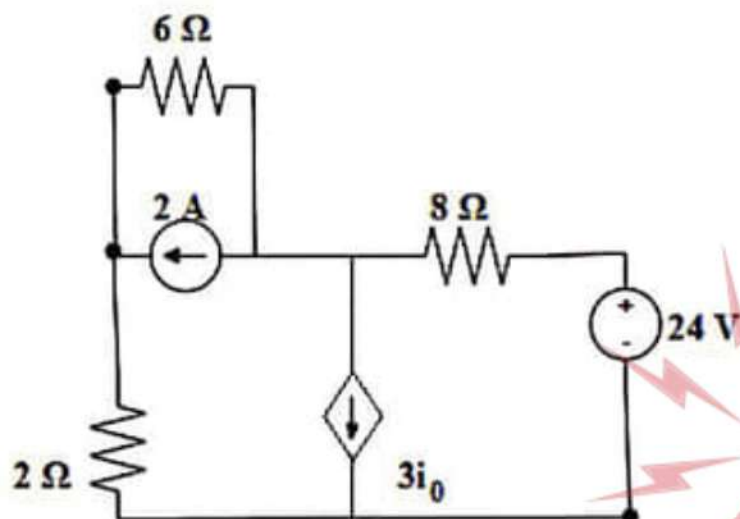


Figure 1.

Q.2 In the circuit of Figure 2:

- Find R so that maximum power is transferred to the resistance R .
- Find this maximum power.

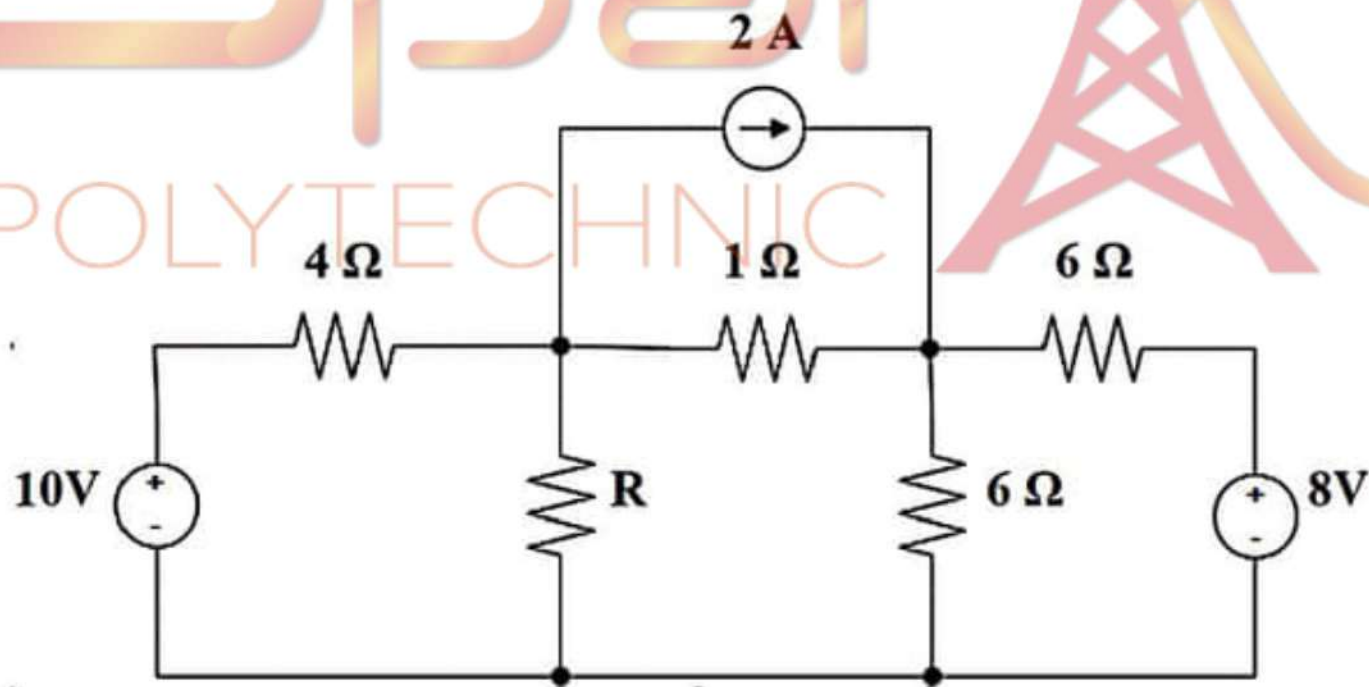


Figure 2.

For the question 1, 2 and 3 consider the circuit shown in figure 3 under dc steady state conditions and answer:

1) The capacitor voltage is:

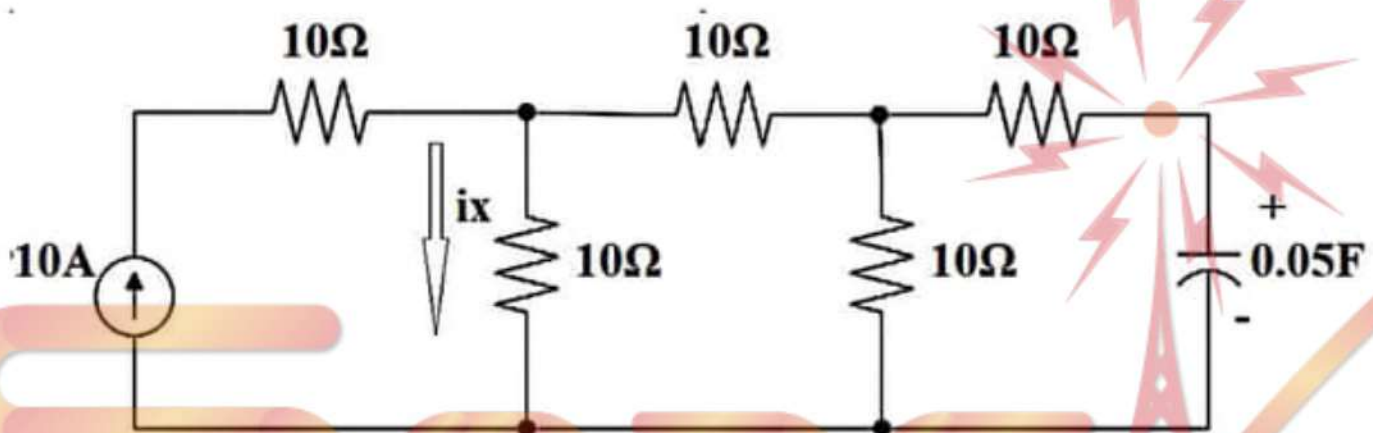
- a. 10V b. 5V c. 20V d. 60V

2) The energy stored in the capacitor is:

- a. 10j b. 5j c. 90j d. 40j

3) The current i_x is:

- a. 12A b. 4A c. 8A d. 3A



For the questions 4, 5 and 6 consider the circuit shown in figure 4. Use Norton's theorem with respect to terminals (a-b) and answer:

4) The Norton current is:

- a. 2A b. 6A c. 3A d. 4A

5) The Norton resistance is:

- a. 0.75Ω b. 3Ω c. 1.5Ω d. 6Ω

6) The current I is:

- a. 1.5A b. 4A c. 3A d. 6A

For the question 7 and 8 consider the current in a 5H inductor is $i(t)=10(1-3t)$ A.

Assume $V(0) = 0$ V. Answer:

7) The voltage at the terminals of the inductor at $t = 3$ s is:

- a. 150V b. -150V c. 350V d. -350V

8) The energy stored in the inductor at $t = 1$ s is:

- a. -320j b. 250j c. 1000j d. 500j

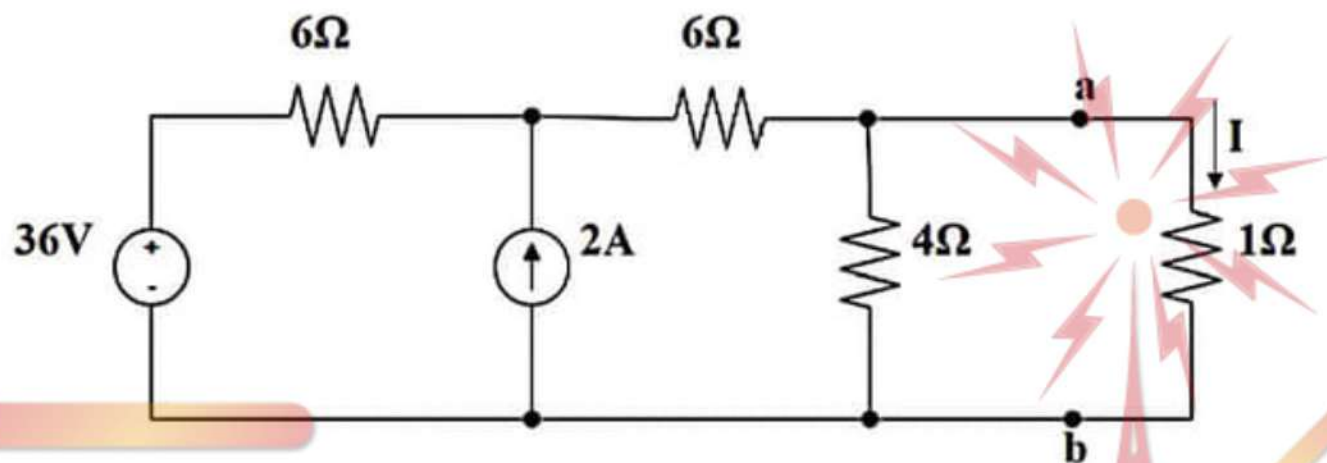


Figure 4.

The switch in Figure 5 has been closed for a long time. It opens at $t = 0$.

9) The steady state response $i(t)$ is

10) The transient response $i(t)$ is

11) The complete response is

12) The inductor current at $t = 200$ ms is

1.5 H

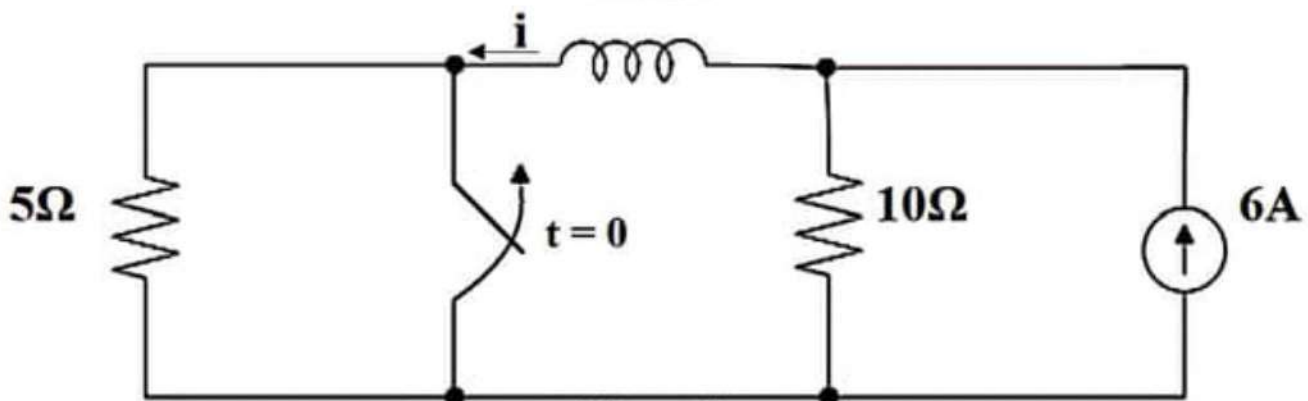


Figure 5.

Q1

1/7/2017

mesh

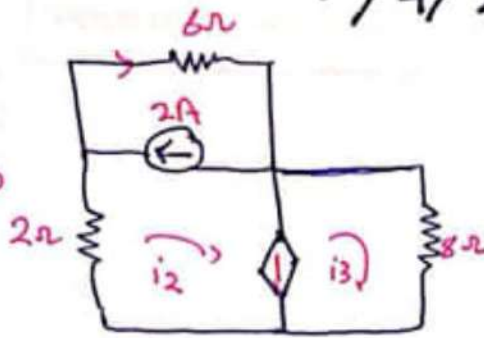
$$6i_1 + 8i_3 + 2i_2 = 0 \quad \text{--- (1)}$$

$$i_1 - i_2 = 2 \quad \text{--- (2)}$$

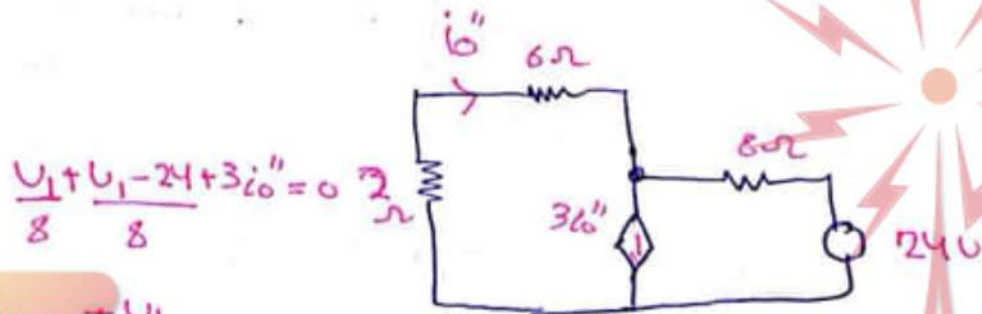
$$i_2 - i_3 = 3i_0' \quad \text{--- (3)}$$

$$i_0' = i_1 \quad \text{--- (4)}$$

$$i_0' = -2.5 \text{ A}$$



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sultanabusharab



$$\frac{U_1}{8} + \frac{U_1 - 24}{8} + 3i_0'' = 0$$

$$i_0'' = -\frac{U_1}{8}$$

$$U_1 = -24 \text{ V}$$

$$i_0'' = 3 \text{ A}$$

$$i_0 = i_0' + i_0'' = -2.5 + 3 = 0.5 \text{ A}$$

$$P_{6\Omega} = i^2 R = 0.5^2 \times 6 = \frac{6}{4} = 1.5 \text{ Watt}$$

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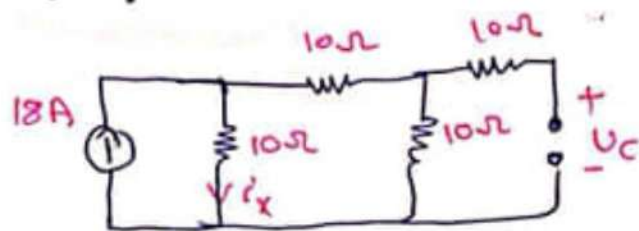
①

Q3) using current divider

$$i_x = 18 \times \frac{20}{10+20} = 12 \text{ A}$$

$$i = 18 - 12 = 6 \text{ A}$$

$$V_c = 10i = 60 \text{ V}$$



$$W_c = \frac{1}{2} C V_c^2 = \frac{1}{2} \times \frac{5}{100} \times 3600 = 90 \text{ J}$$

Q4)

I_N ??

using super mesh

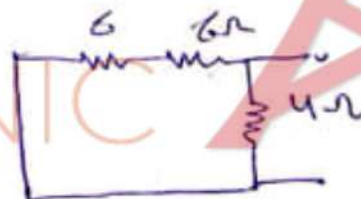
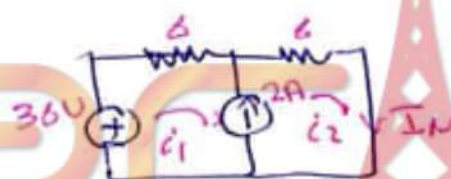
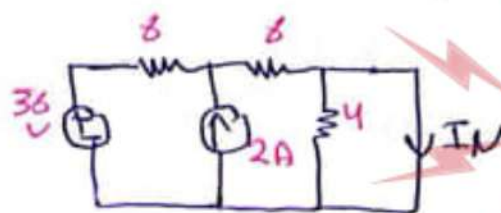
$$6i_1 + 8i_2 = 36$$

$$i_1 + i_2 = 8 \quad \text{--- (1)}$$

$$i_2 - i_1 = 2 \text{ A} \quad \text{--- (2)}$$

$$i_1 = 2 \text{ A} \quad i_2 = 4 \text{ A}$$

$$I_N = i_2 = 4 \text{ A}$$

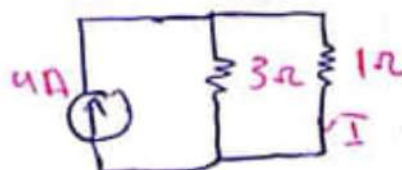


R_N ??

$$6 \text{ series } 6 = 12 \Omega$$

$$12 // 4 = 3 \Omega = R_N$$

$$I = \frac{4 \times 3}{3+1} = 3 \text{ A}$$



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Q5]

$$L = 5 \text{ H}$$

$$i(t) = 10 - 30t \text{ A}$$

$$v(0) = 0 \text{ V}$$

$$v_L = L \frac{di}{dt} = 5 \times \frac{d(10 - 30t)}{dt} = 5 \times -30 = -150 \text{ V}$$

WL at $t=1$

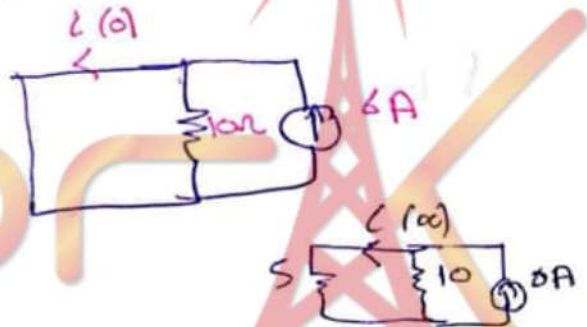
$$WL = \frac{1}{2} Li^2 = \frac{1}{2} \times 5 \times (10 - 30(1))^2 = \frac{1}{2} \times 5 \times 400 = 1000 \text{ J}$$

Q6]

initial condition

$$i(0^-) = i(0^+) = 4 \text{ A}$$

$$i(\infty) = \frac{10}{\frac{10+5}{5}} = 4 \text{ A}$$

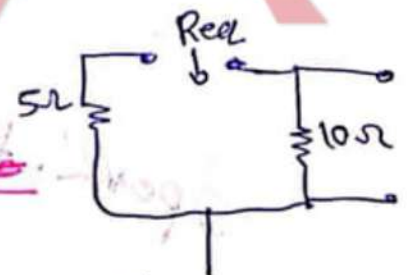


$$i(t) = i(\infty) + (i(0) - i(\infty))e^{-t/\tau}$$

$$i(t) = \left(4 - \frac{4}{2}\right)e^{-t/\tau} = 0$$

$$i(t) = i_0 + i_\infty = 2e^{-t/\tau} \quad t \geq 0$$

$$\tau = \frac{L}{R_{eq}}$$



$$i(200 \text{ ms}) = 2e^{-\frac{0.2}{\tau}}$$

$$\tau = \frac{1 \cdot 5}{15} = 0.1 \quad \dots i = 4 + 2e^{-10t} \text{ A}$$

$$R_{eq} = 5 + 10 = 15$$

$$i(0.2 \text{ s}) = 2e^{-\left[\frac{0.2}{0.1}\right]} = 2e^{-2} = \left(\frac{2}{2.7}\right) 2 \text{ AMP (4)}$$

$$i'(200\mu s) = 4 + 2e^{-10(200\mu s)}$$

$$= 4 + 2e^{-2} = 4.77 \text{ A}$$

$$V(0) = 0 \quad I(0) = 2$$

Q71

Find $\omega, \alpha, V(t)$

$$\alpha = \frac{1}{2R_0} = 12500$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 3162.28$$

$\alpha < \omega_0$ (over damped)

$$V(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$s_1 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -24593.4$$

$$s_2 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -406.8$$

$$V(0) = 0 = A_1 e^0 + A_2 e^0 \Rightarrow A_1 = -A_2 \quad \text{--- (1)}$$

$$\frac{V}{200} + i + C \frac{dV}{dt} = 0$$

$$\frac{V(0)}{200} + i(0) + C \frac{dV(0)}{dt} = 0 \rightarrow \frac{dV(0)}{dt} = \frac{-2}{0.2} = -10 \text{ M}$$

$$\frac{dV(t)}{dt} = A_1 s_1 e^{s_1 t} + A_2 s_2 e^{s_2 t} = -10 \text{ M} \quad \text{--- (2)}$$

$$A_1 = 413.45 \quad A_2 = -413.45$$

$$V(t) = 413.45 e^{-24593.4 t} - 413.45 e^{-406.8 t}$$



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Electrical circuits 1:

Final Exam
Second semester
2017-2018

Date: May. 15, 2018

Exam Time: 100 minuts

Choose the correct answer.

* For the questions 1,2,3,4 consider the circuit shown in figure 1 and answer:

1) The value of V_1 is:

- a) 11.6V b) 22V
c) -22V d) 7.4 V

2) The value of the current (i_0) is:

- a) 2A b) 2.8A
c) 1.4A d) -1.4A

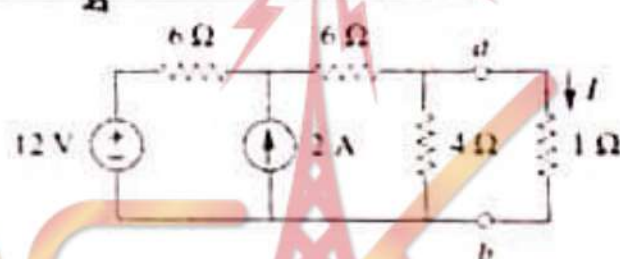
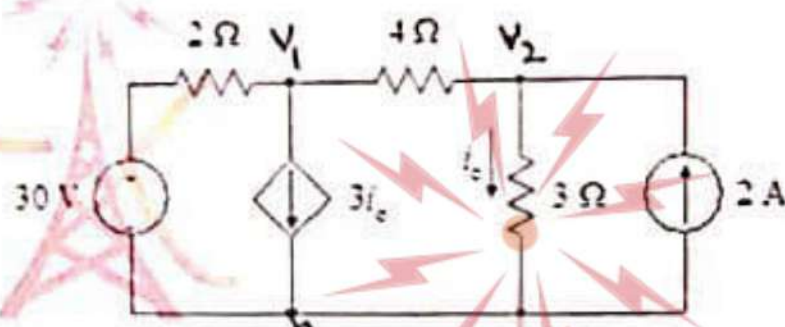
3) The value of the voltage (V_2) is:

- a) -4.8A b) 4A
c) -2.4V d) 8.4v

4) The power absorbed by 4 Ω resistor is:

- a) 6.4W b) 3.64W c) 2.56W
d) 10W

Fig.1



* For the questions 5,6,7 consider the circuit shown in figure 2 :

5) The value of R_{th} with respect to terminals ab is:

- a) 3 Ω b) 6 Ω
c) 2 Ω d) 17 Ω

6) The value of V_{TH} from the terminals (a,b) is:

- a) 3V b) 6V
c) 9V d) 12V

7) The power delivered to 1 Ω resistor is:

- a) 2.25W b) 1.5W c) 4W d) 4.45W

Fig.2

* For the questions 8,9,10,11 consider the circuit shown in figure 3 and answer:

8) The value of $v(t = \infty)$ is:

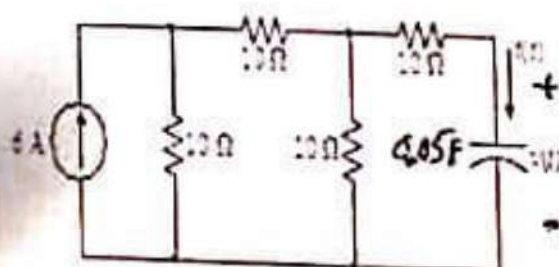
- a) 15V b) -15V
c) 20V d) -20V

9) The value of $i(t = \infty)$ is:

- a) 2A b) 10A
c) 4.68A d) 0A

10) The value of τ (time constant) is:

- a) 1.2s b) 833 ms



c) 1.6 s

d) 8.6 s

Fig.3

11) If The initial value of the capacitor voltage is $v(0) = 10$ V. The initial value of the capacitor current $i(0)$ is:

- a) 0.6 A b) 1.2 A c) 10.63 A d) 3.6 A

* If the voltage across a 1 -mF capacitor is $v(t) = 60 \cos 100t$ mA, answer 12,13:

12) The current through the capacitor is:

- a) $-6 \sin 100t$ mA. b) $6 \cos 100t$ mA
c) $-12 \sin 100t$ mA d) $-6 \cos 100t$ mA

13) The energy stored is:

- a) $1.8 \sin^2(100t)$ μ J. b) $1.8 \cos^2(100t)$ μ J.
c) $1800 \sin^2(100t)$ mJ. d) $1800 \cos^2(100t)$ mJ.

* For the questions 14,15,16,17 consider the circuit shown in figure 3 ($i(t) = 0.6e^{-2t}$ A. If $i(0) = 1.4$ A) :

14) The current $i_2(t)$ is:

- a) 0.8 A b) 1.6 A
c) 6 A d) 3.2 A

15) The total current $i_2(t)$ is:

- a) $(-0.8 + 1.4e^{-2t})$ A b) $(-0.4 + 1.2e^{-2t})$ A
c) $(-0.4 + 1.2e^{-2t})$ A d) $(-0.8 + 1.2e^{-2t})$ A

16) The voltage $v_1(t)$ is:

- a) $(6 + 1.6e^{-2t})$ V b) $(18e^{-2t})$ V
c) $(6 + 1.6e^{-2t})$ V d) $-36e^{-2t}$ V

17) The voltage $v_2(t)$ is:

- a) $(6 - 1.6e^{-2t})$ V b) $(6 + 1.6e^{-2t})$ V
c) $-7.2e^{-2t}$ V d) $(8 + 1.6e^{-2t})$ V

* For the questions 18,19 Refer to figure5:

18) the value of the equivalent resistance is:

- a) 9 Ω b) 4.5 Ω c) 6 Ω d) 4 Ω .

19) the value of the current i_0 is:

- a) 2 A b) 4 A c) 4.5 A d) 8 A

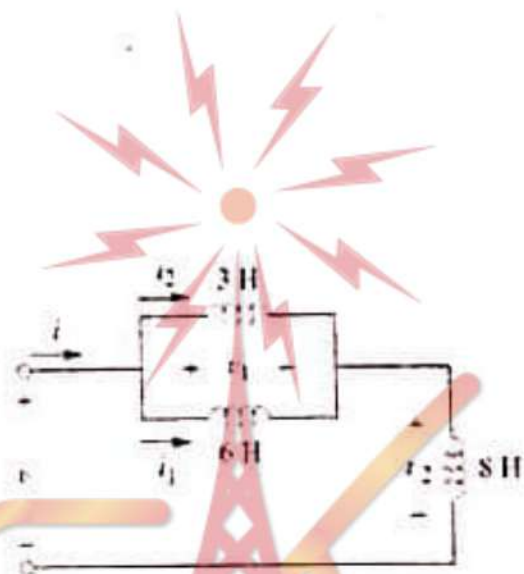


Fig.4



Fig.5

• For the questions 20,21 consider the circuit shown in figure 6 :

20) The voltage across $10\ \Omega$ resistor is:

- a) 4 V b) 50 V
c) 90 d) 45 V

21) The current I_x is:

- a) 5 A b) 2.4 A
c) -2.4 A d) 3 A

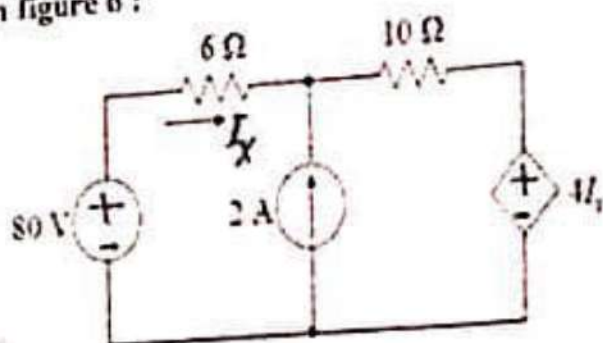


Fig.6

• For the questions 22,23,24 (consider the circuit shown in figure 7. Assuming $v(0) = 5V$, $i(0) = 0$, $L = 1H$, $C = 10mF$. And $R = 1.923\Omega$)

22) The value of α (damping factor) is:

- a) $13\ s^{-1}$ b) $52\ s^{-1}$
c) $26\ s^{-1}$ d) $54\ s^{-1}$

23) The value of ω_0 (resonant frequency) is:

- a) $52\ s^{-1}$ b) $10\ s^{-1}$
c) $20\ s^{-1}$ d) $26\ s^{-1}$

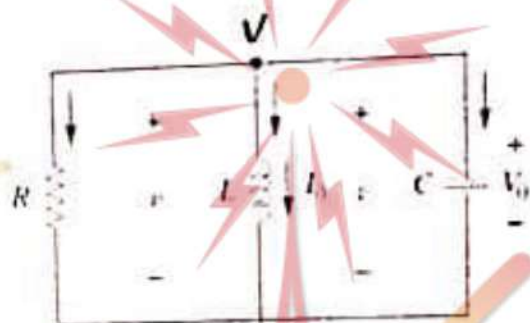


Fig.7

24) The general natural response $v(t)$ for $t > 0$ is:

- a) $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ b) $v(t) = e^{-\alpha t} (A_1 t + A_2)$
c) $v(t) = e^{-\alpha t} (A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t})$ d) $v(t) = e^{-\alpha t} (B_1 \cos\omega_d t + B_2 \sin\omega_d t)$

25) A heater draws 10 A when connected to 240V supply. How long does it take to consume 480 kJ.

- a) 260 ms b) 5m s
c) 20s d) 200 s

1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	

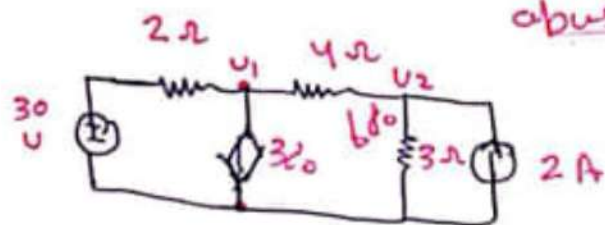
Good Luck

فاینل برین ۱

15/3/2018

suftan
abushanab

① 11.6V (A)



② $i_o = \frac{8.4}{3} = 2.8A$ (B)

③ 8.4 (D)

④ $P = \frac{U^2}{R} = U \cdot i = i^2 R$
 $= \frac{(11.6 - 8.4)^2}{4} + 4 = 2.5W$ (C)

$$\frac{U_1 - 30}{2} + 3i_o + \frac{U_1 - U_2}{4} = 0$$

$$i_o = \frac{U_2}{3}$$

$$2U_1 - 60 + 4U_2 + U_1 - U_2 = 0$$

$$3U_1 + 3U_2 = 60 \quad \text{--- (1)}$$

$$\frac{U_2 - U_1}{4} + \frac{U_2}{3} = 2$$

$$3U_2 - 3U_1 + 4U_2 = 24$$

$$-3U_1 + 7U_2 = 24 \quad \text{--- (2)}$$

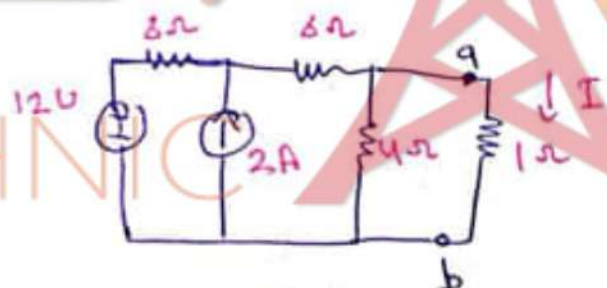
$$U_1 = 11.6V$$

$$U_2 = 6.4V$$

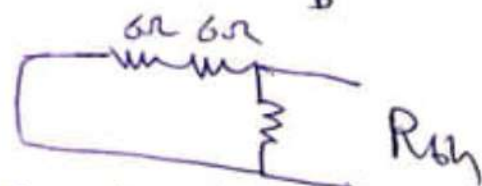
#spark-team

⑤ 3Ω (A)

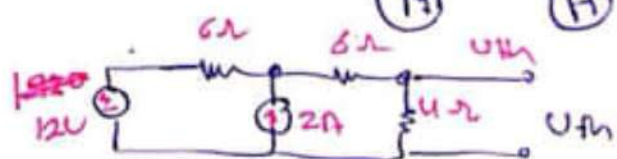
⑥ 6V (B)



⑦ $U_1 = 1 + 0.5 = 1.5V$
 $P = \frac{U^2}{R} = \frac{(1.5)^2}{1} = 2.25W$ (A)



$$R_{th} = (6 + 6) \parallel 1 = 3\Omega$$



$$2 = \frac{U - 12}{6} + \frac{U - U_{th}}{8}$$

$$20 - U_{th} = 24$$

$$\frac{U_{th}}{4} + \frac{U_{th} - U}{8} = 0$$

$$6U_{th} + 4U_{th} - 4U = 0$$

$$-4U + 10U_{th} = 0$$

$$U_{th} = 6V$$

(P)

⑧ 20 V (C)

⑨ zero A (D)

⑩ 833 ms (B)

⑪ $V(0) = 10 \text{ V}$

$$V(t) = 20 + [10 - 20] e^{-t/63 \cos}$$

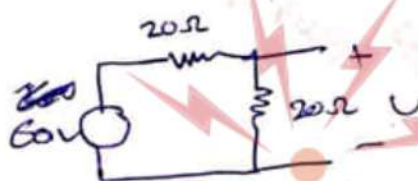
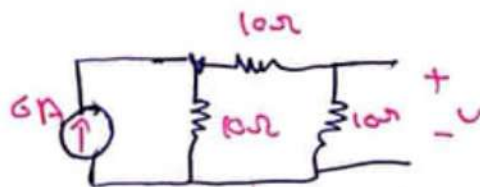
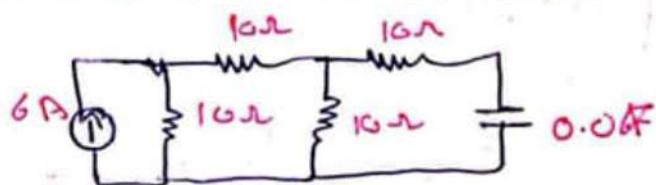
$$= 20 - 10 e^{-12t}$$

$$i = C \frac{dv}{dt}$$

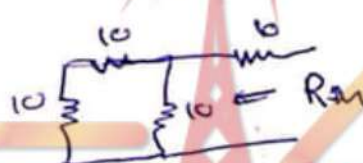
$$i = 0.05 \times 12 e^{-12t}$$

$$i = 0.6 e^{-12t}$$

$$i(0) = 0.6 \text{ A} \quad (\text{A})$$



$$V = \frac{60 + 60}{30} = 20 \text{ Volt}$$



$$R_{th} = (10 \parallel 10) \parallel 10 + 10$$

$$= 16.66 \Omega$$

$$\tau = 0.05 \times 16.66$$

$$= 833 \text{ ms}$$

⑫ $C = 1 \text{ mF}$ $v(t) = 60 \cos 100t \text{ mV}$

$$i = C \frac{dv}{dt}$$

$$= 0.01 \times (-6000 \sin 100t)$$

$$= -6 \sin 100t \text{ mA} \quad (\text{A})$$

⑬ $W = \frac{1}{2} C V^2$

$$= \frac{1}{2} \times 0.01 \times (60 \cos 100t \text{ mV})^2$$

$$= 1.8 \cos^2 100t \text{ mJ}$$

(B)

(2)

(14)

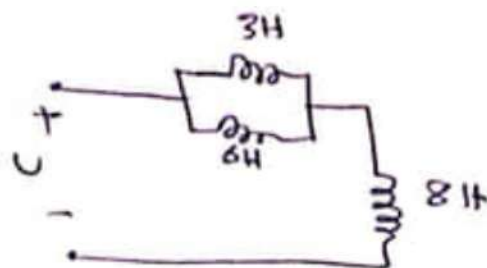
$$i(t) = 0.6 e^{-2t}$$

$$i(0) = 1.4 \text{ A}$$

$$i'(0) = i'(1) + i'(2)$$

$$1.4 = 0.6 + i_2'(0)$$

$$i_2'(0) = 0.8 \text{ A} \quad \textcircled{A}$$



(15)

$$U_{6H} = U_{3H}$$

$$U = L \frac{di}{dt} = 6 + 0.6 + -2e^{-2t} = -7.2 e^{-2t}$$

$$i_2(t) = \frac{1}{L} \int_0^t U dt + i_2(0)$$

$$= \frac{1}{3} \int_0^t -7.2 e^{-2t} dt + 0.8$$

$$= \frac{1}{3} \left[\frac{7.2}{-2} e^{-2t} \right]_0^t + 0.8$$

$$= 1.2 e^{-2t} - 1.2 + 0.8$$

$$= 1.2 e^{-2t} - 0.4 \quad \textcircled{B}$$

(16)

$$i(t) = i_1(t) + i_2(t)$$

$$= 0.6 e^{-2t} + 1.2 e^{-2t} - 0.4$$

$$= 1.8 e^{-2t} - 0.4$$

$$U = L \frac{di}{dt} = 10 + 1.6 + -2e^{-2t}$$

$$= -3.6 e^{-2t} \quad \textcircled{D}$$

(17)

$$U_{6H} = U_{3H}$$

$$U = L \frac{di}{dt} = 6 + 0.6 + -2e^{-2t}$$

$$= -7.2 e^{-2t} \quad \textcircled{C}$$

(18)

$$R_{eq} = ((20 + 47) // 3.53) + 1.411$$

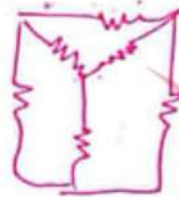
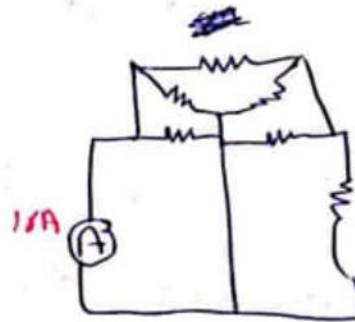
$$R_{eq} = 4.5 \Omega \quad (B)$$

$$3.0886 \Omega + 1.411 = 4.5 \Omega$$

$$20 + 4.7$$

$$= 24.7$$

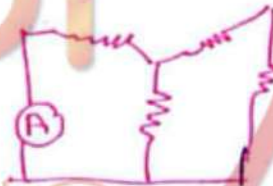
$$24.7 // 5.53 = 3.0886$$



$$R_1 = \frac{6 + 8}{34} = 1.411 \Omega$$

$$R_2 = \frac{20 + 6}{34} = 4.7 \Omega$$

$$R_3 = \frac{30 + 6}{34} = 3.53 \Omega$$



(19)

$$V_{3.0886} = 3.0886 + 16$$

$$= 49.4776 V$$

$$I_{24.7} = \frac{49.4776}{24.7} = 2 A$$

(A)

POLYTECHNIC

Sultana
Abushehah

S per R team

(4)

20

$$i_x = 4$$

#spark-team

Sultan
Abushehab

$$-80 + 6i_1 + 10i_2 + 4i_1 = 0$$

$$6i_1 + 10i_2 + 4i_1 = 80$$

$$10i_1 + 10i_2 = 80 \dots (1)$$

$$i_1 + 2 = i_2$$

$$i_1 - i_2 = -2 \dots (2)$$

$$i_1 = 3A$$

$$i_2 = 5A$$

$$i_x = 3A$$

$$(20) \quad V_{20} = 5 \times 10 = 50V \quad (B)$$

$$(21) \quad i_x = 3A \quad (D)$$

22

$$V(0) = 5V, \quad L = 7H$$

$$i(0) = 0$$

$$C = 10\mu F$$

$$R = 1.923\Omega$$

$$\chi = \frac{1}{2Rc} = \frac{1}{2 \times 1.923 \times 10^{-5}} = 26s^{-1} \quad (C)$$

$$(23) \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.01}} = 10s^{-1} \quad (B)$$

24

$$\omega > \omega_0$$

$$V(t) = A_1 e^{-\omega t} + A_2 e^{+2t} \quad (A)$$

25

$$i = 10A$$

$$V = 240V$$

$$\omega = 480KJ$$

$$P = i \times V = 10 \times 240 = 2400W$$

$$\omega = P \times t = 480 \times 10^3 = t \times 2400 \quad t = 200s \quad (D)$$

5